Characterizing Canadian Out-of-district Political Donations from 2015 to 2024*

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Federal political representation in Canada is geographical: one Member of Parliament (MP) represents one electoral district. Despite this representational structure, Canadians can legally donate to a host of political entities, regardless of proximity. Using a public dataset from Elections Canada's Open Data on Contributions matched with census data from Statistics Canada, we characterize broad patterns in Canadian out-of-district donation from 2015 to 2024. We find that out-of-district donations form a persistent share of donor-sourced funding to both candidates and district associations, averaging 40% across time, election periods, and party lines. While relatively constant through time, the propensity for out-of-district donation varies substantially across districts, beyond what spatial proximity alone would predict. Districts with higher population density, median household income, educational attainment, and visible minority proportions see higher rates of out-of-district giving, the same traits that predict donor prevalence generally in Canada. Despite its frequency, out-of-district donation in Canada remain largely regionalized; nearly half of donations that leave their district are received by a political entity in a neighbouring one, and few cross provincial lines. These and other findings provide a helpful characterization of this largely undocumented facet of Canadian political financing, and inform further research on effective representation in Canada and other nations with similar representational structures.

Keywords: out-of-district donation; Canadian politics; political geography; political financing

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1 Introduction

In Canada, federal representation is structurally geographical; citizens are grouped into a few hundred electoral districts based on where they live, each represented by a single Member of Parliament (MP). An individual's political influence in these systems is canonically exercised by voting for their parliamentary candidate of choice during election periods, however alternate forms of engagement exist to support a broader scope of political activities, political entities, and without the spatio-temporal constraints imposed by elections. Citizens may volunteer at their local district associations, advocate for their preferred national party online, vote in intra-party nomination contests, or contribute financially (i.e. donate) to a candidate's campaign (Bednar and Gerber 2011; Baker 2016; Garnett 2024).

Whether intentional or not, many of these alternative forms of political participation can serve to extend a citizen's political influence beyond their district's boundaries. Among these, out-of-district donations are perhaps the clearest and most flexible mechanism through which individuals can influence political entities and activities outside their district, for two main reasons. First, donations are largely unconstrained in time and space. In North America, citizens and permanent residents can legally contribute to various federal political entities at any time, in any district, and as often as they want, subject to annual and per-election limits (Gimpel et al. 2008; Elections Canada 2026a). Second, in North America, contributions from individuals in general are the primary source of funding for electoral campaigns (Gimpel et al. 2008; Garnett 2024; Elections Canada 2026f) and a major source of revenue for national parties (Corrado et al. 2010; Garnett et al. 2022). In the U.S., even incumbent representatives often attribute a majority of their income to their donor base (Baker 2016). While the activities of many of these entities may be reimbursed through public stipends and subsidies, the gross funding needed for a party leadership contestant or electoral candidate to run a strong campaign often needs to available on-demand, and is expected (Garnett 2024). Despite a growing body of Canadian research documenting the extent of and patterns of federal political donations generally (e.g. Besco and Tolley 2022; Tolley et al. 2022; Garnett 2024), comparatively little attention has been given to out-of-district donations.

Existing research on out-of-district federal contributions in the U.S. documents substantial rates of out-of-district donation. Both congressional and senatorial candidates presently source between 50 and 70 percent of their campaign funding from outside their district, a proportion that has increased in recent decades (Gimpel et al. 2008; Bednar and Gerber 2011; Sievert and Mathiasen 2023). Incumbent representatives also receive around half of their donations from out-of-district, which meaningfully influences their roll-call behaviour (Peoples and Gortari 2008; Canes-Wrone and Miller 2022). These donations are thought to be primarily motivated by partisan gain, evidenced, for example, by increased donations to districts with close races (Gimpel et al. 2008; Jacobs and Imboywa 2024). Both major parties in the U.S. primarily source local funding from a single small group of highly influential and socio-demographically similar districts (Gimpel et al. 2006).

In this paper, we document the extent of, and large-scale trends in, Canadian out-of-district donations to electoral candidates and district associations from 2015 to 2024. We use a database containing all contributions received by federal political entities during this period, made publicly accessible by *Elections Canada*. This dataset was augmented by situating donors geographically, and matching donations to their socio-demographic environment using the 2021 Census, allowing us to better understand which factors correlate with out-of-district donation rates. A notable by-product of our research process was the development of a reproducible workflow for acquiring, cleaning, augmenting, and validating a dataset of donor-geolocated individual contributions to all federal political entities tracked under the *Canada Elections Act* from 2015 to 2024[1].

We find that, between 2015 and 2024, out-of-district donations account for a persistent share of donor-sourced funding to both electoral candidates and district associations, averaging 40% across time, election periods, and party lines. This relative invariance does not extend to space, however: the propensity for out-of-district donation changes substantially from district to district. Donations originating from densely populated, high-income, and well-educated districts are more likely to leave their district of origin, echoing predictors of out-of-district giving established in the U.S. Our analysis also shows that the visible minority proportion of a donor's district is the most correlated socio-demographic variable that we considered. This finding departs from the ethnic profile historically associated with "donor districts" in the U.S. (Gimpel et al. 2008; Canes-Wrone and Miller 2022), but aligns with recent Canadian research linking minority ethnic communities to higher rates of political participation and out-of-district giving (Garnett et al. 2022; Besco and Tolley 2022). Despite the considerable presence of out-of-district donations during our study period, out-of-district giving remains a regionalized phenomenon: nearly half of all donations that leave their district of origin are addressed to political entities in a neighbouring one, and fewer still cross a provincial boundary, limiting their potential influence on geographically-based representation. Although we do not formally evaluate the representational consequences of out-of-district donations, the systematic characterization presented here provides a foundation for future research on political finance and effective representation in Canada, while also advancing understanding of how political influence is distributed across the electorate.

[1]Data acquisition and processing procedures are available at 10.5281/zenodo.21708657.

2 Background

2.1 Motivation: Representational Effects of Out-of-district Donations in the U.S.

The extent, trends, and socio-demographic characteristics of U.S. out-of-district donations are well- documented among individual contributions to campaigning and incumbent House representatives (both senators and congressional representatives). This body of research was catalyzed by Gimpel et al. (2008)'s examination of out-of-district donations during four Congressional elections between 1994 and 2004, and their impacts on legislative representation. They propose the existence of "donor districts"; a small set of older, wealthy, well-educated, and urban districts wherein a large majority of out-of-district donations originate, destined to Republican and Democrat candidates alike. Notably, these socio-demographic predictors of out-of-district donor output are largely the same as predictors of donor output generally in both the U.S. and Canada (Gimpel et al. 2006; Garnett 2024). Over this time period, they also observed an increase in the proportion of non-local donations received by the average candidate, from around 50% in 1994 to nearly 70% in 2004 (Gimpel et al. 2008). More recent studies suggest that this and similar trends have continued past 2004 (Bramlett et al. 2011; Baker 2016; Canes-Wrone and Miller 2022), and for donations made during U.S. senate elections (Sievert and Mathiasen 2023) Driven by the observation that the "closeness" of local races strongly predicted which districts received these non-local donations, Gimpel et al. (2008) argue that gaining a partisan advantage (and not seeking ideological or identity-based representation) is the primary motivation behind non-local donations.

Follow-up U.S. research has found that out-of-district campaign contributions affect the voting behaviour of Congressional representatives, thereby distorting the supposedly-dyadic relationship between a representative and their district. A study of the 2006-2010 Congressional elections revealed that the average congressional representative becomes less responsive to their constituencies' ideological preferences with increased out-of-district contributions (Baker 2016). Canes-Wrone and Miller (2022) further explore the 2006-2016 Congressional elections to show that House voting behaviour is indeed associated with the preferences of their national donor base, contrasted against research finding that political action committees do not affect roll call behaviour (Ansolabehere et al. 2003). Another study on senatorial elections emphasizes the rivalrous relationship between constituents and out-of-district donors, experimentally showing that constituents who learned about outside donation expected their senators to be less loyal to local interests (Nathan et al. 2025). More generally, longitudinal analyses of out-of-state (and therefore out-of-district) donations during senatorial elections echo many of the findings from congressional elections (Sievert and Mathiasen 2023; Jacobs and Imboywa 2024; Keena 2024).

Strikingly, out-of-district federal political donations have not been analyzed to nearly the same extent in other Westminster-style parliamentary democracies, including Canada. Besides differences in population size and the scale of electoral competition, we propose two barriers to research in the Canadian context. First, U.S. annual contribution limits are much higher than Canadian ones, but the threshold of location disclosure is the same (Gimpel et al. 2006; Federal Election Commission 2026; Elections Canada 2026f). This means that location-disclosed donations (and therefore those that can be analyzed as out-of-district) likely constitute a larger share of the total amount of all donor-based federal political financing. Second, the data manipulation steps required to situate donors within districts and attribute accurate Census-based data to the entries are more straightforward in the U.S. as compared to Canada. U.S. Census information is available directly at the zip-code level, and their geographic conversion files (to situate zip-codes within districts) are fully open-source. While donor geolocation and Census-based data augmentation has been done in some Canadian research (e.g. Garnett 2024), most of the Canadian donation studies we identified rely on political surveys such as the Canadian Elections Study.te.

2.2 Brief Review of Donations to Canadian Electoral Candidates

Limited accounts of out-of-district donation rates exist in the Canadian context. One study analyzing co-ethnic affinity among donors and candidates found that the likelihood of a donor donating out-of-district is higher for ethnic minorities (50-70%) compared to donors of European origin (33%) (Besco and Tolley 2022). However, these proportions only describe candidate-bound donations during the 2015 general election period. Here we review existing research concerning donations to Canadian electoral candidates, contextualizing our subsequent analysis[2]. We do not consider studies focusing on donations to other political entities as these are outside the scope of our study[3].

Considerable research has examined socio-demographic characteristics of donors and candidates in Canada. Preliminary analyses of individual donations from 2006 to 2011 (Jansen et al. 2012; Carmichael and Howe 2014), and a more recent analysis of candidate-bound donations during the 2015, 2019, and 2021 general elections (Garnett 2024) have found that donors are more likely to originate from older, wealthier, and better-educated districts. The gender gap in giving has also received attention, likely given the broader context of established gender gaps in political

[2]Although donations to district associations are also considered in our study, we were unable to find research focusing on contributions addressed to these.

[3]We refer the reader to Young and Jansen (2011), Garnett et al. (2022), and Pruysers (2024) for examples of such studies.

participation (Tolley et al. 2022; McMahon et al. 2023). From 2004 to 2015, overall federal donation amounts have increased, particularly in Western Canada, and among Conservative political entities (McMahon and Wolfe 2015). However, contribution rates to candidates during the following three general elections (in 2015, 2019, and 2021) have since decreased (Garnett 2024), and the COVID-19 pandemic saw overall reduced political participation (McMahon et al. 2023).

Canadian donors tend to donate more to candidates that they share aspects of their identity with. For instance, women donors are more likely to donate to women candidates (Tolley et al. 2022), and there are high rates of co-ethnic donation, particularly among South Asian donors (Besco and Tolley 2022). More generally, South Asian political participation appears to be particularly heightened in Canada, relative to other ethnocultural groups (Garnett 2024).

Some studies have also attempted to assess the impacts of political donations on representation. Among incumbent legislators, voting behaviour is seemingly independent of the composition of their donor base (Peoples and Gortari 2008). However, Eagles (2004) found that local campaign donations during the 1993 and 1997 elections meaningfully affected the number of votes a candidate receives.

3 Data and Methods

3.1 Data Overview

The data used in this paper was originally sourced from the Elections Canada Open Data on Contributions datasets (Elections Canada 2025a). Since 2004, the *Canada Elections Act* has required that all contributions (monetary or otherwise) valued at over $20 be recorded by the recipient and shared with the government (Elections Canada 2026f). The Elections Canada dataset on donations contains all donations made by individuals to registered federal political entities since 2004 (Elections Canada 2025a). Additionally, donors who contributed over $200 must have their full name and address (including their postal code) recorded (Elections Canada 2026f). For all contributions, Elections Canada also records the recipient's name, affiliated district (if applicable), party (if applicable), entity (e.g. national party, candidate, etc.), and the date the contribution was received. More information on political financing laws in Canada is presented in Appendix A. The primary dataset used in this study is a cleaned version of Elections Canada's, made available to us by the Investigative Journalism Foundation. Their cleaning process is described in Appendix B.

To even be considered as out-of-district, a federal donation must be received by a political entity that is representationally tied to a specific district. For this reason, only donations received by candidates during election periods and district associations[4] (hereafter referred to as localized entities) can be considered in the study of out-of-district donations. Donations received by non-localized entities present in this dataset (parties or leadership contestants) cannot. Given their prevalence, however, we briefly compare these with localized entities in Section 3.2.1.

Every ten years, federal electoral district (FED) boundaries are redrawn to account for population changes that occurred within the decade (Elections Canada 2025b). To restrict our focus to a single, static set of FEDs, our analysis only includes contributions made during the 2013 representational order, effective between October 2, 2015 and April 22, 2024 (Elections Canada 2025b). This is the largest such order present in the dataset, capturing 61% of all recorded contributions since 2004, and spanning three general election periods (42nd, 43rd, and 44th). Donors were matched to their FEDs by their postal codes using Statistics Canada's Postal Code Conversion Files[5] [PCCF; Statistics Canada (2025)], made available to us by the University of Toronto. Donor FEDs were then compared with recipient FEDs to assess whether a donation was out-of-district. Donor location was not recorded for the majority of donations under $200, and the small share that could be donor-geolocated were distributed very differently from the rest of the data. For these reasons, this study only considers donations above $200, as was done in similar studies (Gimpel et al. 2008; Garnett 2024). These smaller donations are discussed further in Appendix D.

Donations were also situated within census dissemination areas (DA), allowing us to attribute socio-demographic characteristics to donation sources at a fine spatial grain. For each dissemination area, we identified its population, population density, median household income, mean age, proportion of the population with a post-secondary education, and proportions of various visible minority ethnicities. A full list of the ethnicities is available in Appendix C. These variables were chosen since they are associated with the propensity to donate during an election (Gimpel et al. 2006, 2008; Garnett 2024). Further, ethnicities were disaggregated since previous Canadian research has found that some minority groups such as South Asians

[4]Although nomination contestants and by-election candidates are also local political entities and were present in the data, these were not considered to simplify our analysis and discussion, since they only make up 0.7% of donations addressed to localized entities.

[5]Postal codes may straddle multiple FEDs. For this study, we created a deterministic spatial lookup table that resolves these cases using reasonable criteria. Donations whose postal codes could not be matched in this way were dropped. Our methodologies, and statistical assessments of their impacts are detailed further in Appendix C.1.

donate at much higher rates than others (Besco and Tolley 2022). We again used the PCCF to match donor postal codes to the DAs of the 2021 Census, and to group DAs into FEDs[6] to deduce the same socio-demographic information at the district level. A detailed walkthrough of the full data pipeline, from the files available for download on Canadian government websites to the core datasets used for analysis can be found in Appendix B and Appendix C. All data cleaning, visualizations, and analyses were conducted using the R statistical software (R Core Team 2025), and viewable at 10.5281/zenodo.21708657.

3.2 Data Summaries

3.2.1 Situating Localized Donations

We first provide preliminary summaries of all contributions[7] during our study period, to help situate further analysis of out-of-district donations. Table 1 breaks down donations by recipient type. National entities (parties and leadership contestants) received the majority (80.8%) of contributions during our study period. Among donations addressed to localized entities, 24.3% are received by candidates, and 75.7% were received by district associations (EDAs)[8]. Proportions are similar when broken down by count instead of amount.

Figure 1 shows how contributions during our study period are distributed in time[9], for national and localized entities separately. Among national-level donations, we observe three distinct temporal signatures. First, donation rates are higher in December of every year during our study, likely since chartiable donations tend to be more common in this period in general, as individuals claim tax deductions and give more during the holidays. Second, we observe six peaks corresponding to general election periods (peaks during and immediately before grey shaded regions), and Conservative party leadership campaigns (peaks in the blue trend line in the first half of 2017, 2020, and 2022). Third, national-entity donation rates generally increased over our study period, following the increasing trend observed from 2004 to 2015 by McMahon

[6]Dissemination areas may straddle multiple FEDs. Appendix C.1 discusses how these cases were handled.

[7]Throughout this paper, the terms "contribution" and "donation" are used interchangeably and implicitly refer to contributions above $200 that were successfully donor-geolocated, unless otherwise specified.

[8]In our dataset, candidates receive all of their donations during writ-periods. During these periods, they instead represent 48% of the total amount contributed to localized entities, with district associations making up the remaining 52% (Appendix G)

[9]All donations to electoral candidates and leadership contestants were tied to specific events in time. A small proportion (0.7% of the total contributed amount) of contributions to district associations and national parties over $200 were missing specific dates, but could always be attributed to a fiscal year. Plots over time in this report include these cases, which slightly exaggerates the observed December peaks in contribution rates.

Table 1: Summaries of financial contributions to federal political entities over $200 from 2015 to 2024, separated by entity. Only localized contributions (those addressed to candidates and district associations) are considered further in our study of out-of-district donations. Contributions received by nomination contestants are not shown, nor studied.

Political Entity	Type	Donation Count	Donation Amount (Thousands)	Share of Total Amount
Candidate	Localized	30,559	$19,883	4.6%
District Association	Localized	108,619	$63,354	14.6%
Leadership Contestant	National	47,873	$32,451	7.5%
Party	National	618,609	$319,056	73.4%

and Wolfe (2015) during the previous decade. This is particularly noticeable after the 2021 general election, and for donations received by Conservative national entities.

Donation rates to localized entities exhibited the same December peaks, and peaks during general election periods that were observed for donations to national entities. However, locaized donation rates did not exhibit an increasing trend, unlike their counterparts bound for national entities. We also observe a significant reduction in donation rates from the start of 2019 to mid-2021, which may be partially explained by the COVID-19 pandemic, agreeing with previous findings on reduced contribution rates during this period (McMahon et al. 2023). Spatial comparisons between the aggregated donations bound for national vs. localized entities during the study period yielded only minor differences, and are presented in Appendix G, which did not yield drastic differences.

3.2.2 Spatio-temporal Trends in Out-of-District Donation Rates

In this section, we present data summaries describing when and where out-of-district contributions are more or less common. Table 2 displays the proportion of contributions above $200 (raw and amount-weighted) that were sent out-of-district, for candidate- and EDA-bound donations. Overall, candidates received 37.6% of these donations from outside their district, and EDAs received 43.8%[10]. In both cases, these proportions increase when donations are

[10] A comparison of the value distributions of in- and out-of-district donations is available in Appendix G.

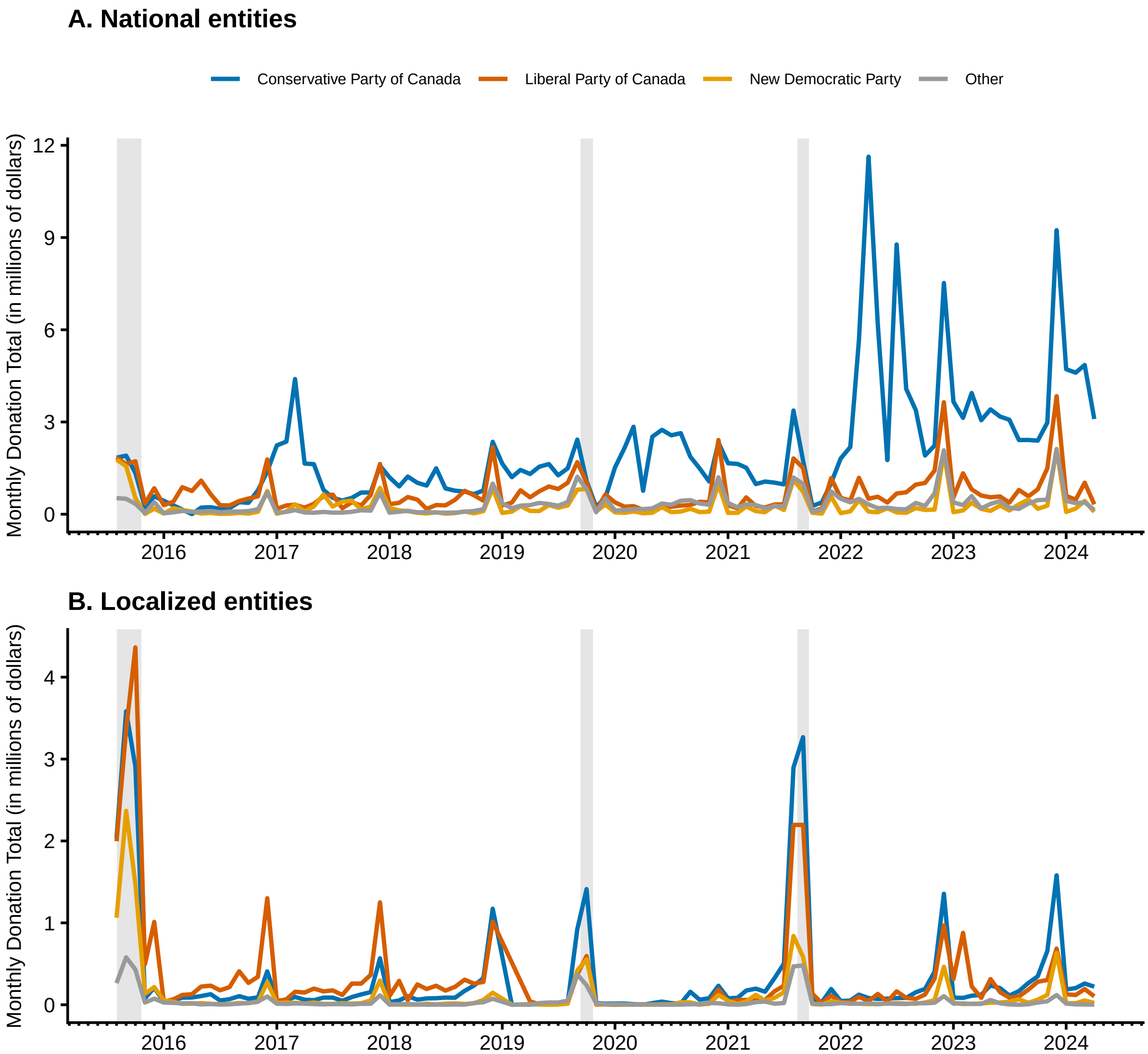


Figure 1: Timeline of financial contributions to federal political entities over \$200, from 2015 to 2024. (A) shows only contributions to national entities (parties and leadership contestants), and (B) shows only contributions to localized entities (candidates and district associations). Within each panel, trendlines are separated by the party of the recipient. Shaded vertical bars indicate the 42nd, 43rd, and 44th general election periods respectively. Graphs were qualitatively similar when contributions under \$200 were included.

Table 2: Proportion of contributions over $200 that were received by localized entities (candidates and district associations) outside the donor's district, from 2015 to 2024. Raw and value-weighted proportions of out-of-district donations (OOD) are reported. Overall contribution counts and amount are also reported for reference. Contributions received by candidates outside general election cycles (by-elections) are not included in calculations.

Political Entity	Donation Count	Donation Amount (Thousands)	OOD Proportion	OOD Proportion (Amount Weighted)
Candidate	30,559	$19,883	37.6%	40.2%
District Association	108,619	$63,354	43.8%	47.8%

weighed by their value, indicating that out-of-district donations are larger, on average, than their within-district counterparts.

Table 3 and Figure 2 show how out-of-district funding rates change over time, for candidates and district associations respectively[11]. Candidate out-of-district rates show a slight decreasing trend, from 39.5% in the 42nd general election in 2015 to around 35.4% in subsequent periods (Table 3). However, this may reflect the heightened absolute number of contributions garnered in 2015 compared to other periods. EDAs received roughly the same proportion of out-of-district contributions during the entire study period, including during writ periods[12].

Figure 3 displays the relationship between in- and out-of-district contribution amounts, for each electoral district present in our dataset. We display donations to candidates and district associations separately. In both cases, out-of-district contribution rates vary greatly from district to district[13]. Some Toronto metropolitan districts such as Toronto—St. Paul's and University—Rosedale sent between 70% and 80% of their contributions to localized entities in other districts, while others such as Alberta's Lethbridge and Ontario's Niagara Falls only sent between 5% and 15% of their localized contribution sum to other districts. Many districts, especially for donations addressed to candidates, have extremely low out-of-district donation rates, as illustrated by the line of points near $y = 0$ in Figure 3. We also find that, at least among the few top districts by contribution amount, out-of-district propensity is not a simple

[11]It is unclear why donations to EDAs plummet in 2019 in our dataset. This does not affect our analysis however, since we never use dates directly.

[12]Roughly similar rates, and the same consistency through time, were observed when entities from individual parties were examined (Appendix G).

[13]The marginal distribution of out-of-district proportions across all districts is shown in Appendix G.

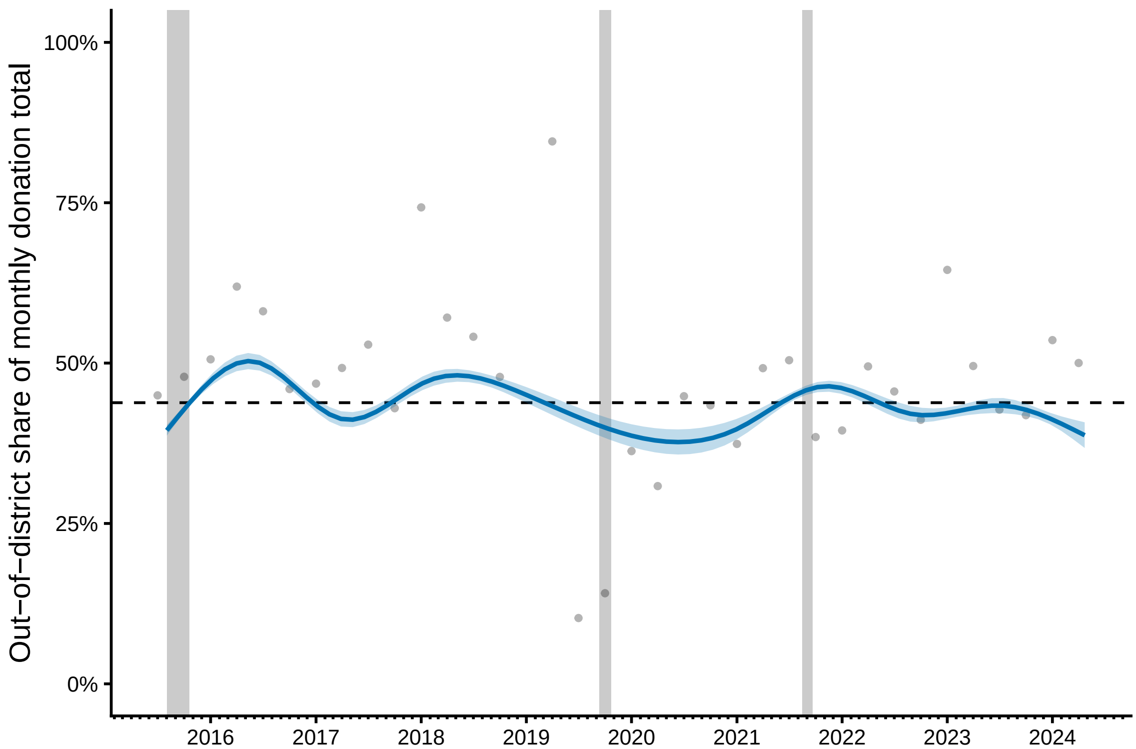


Figure 2: Timeline of the out-of-district share of the total monthly contributed amount over $200 addressed to district associations from 2015 to 2024. Raw monthly proportions are displayed as grey points, and the fitted trend over time (with its 95% confidence interval) is shown with the blue line and ribbon. Trend over time was fit using a generalized additive model with a smooth term over donation date, estimated at the individual level. This is shown instead of a simple line plot to account for the large month-to-month variation in donation rates to district associations generally. Shaded vertical bars indicate the 42nd, 43rd, and 44th general election periods respectively. The horizontal dashed line represents the overall share of contributed amount that was received from out-of-district during the study period, for reference.

Table 3: Proportion of contributions to electoral candidates over $200 that were out-of-district (OOD) over the three general election periods that occurred during our study period. Raw and value-weighted proportions are reported, along with the overall number of contributions (in- and out-of-district) for reference. Contributions received by candidates outside general election cycles (by-elections) are not included in calculations.

General Election	Donation Count	Donation Amount (Thousands)	OOD Proportion	OOD Proportion (Amount Weighted)
42nd general election	16,662	$10,383	39.5%	41.9%
43rd general election	7,900	$5,204	35.2%	39.1%
44th general election	5,997	$4,296	35.6%	37.7%

function of urbanization[14]; Vancouver Granville in British Columbia and Brampton North are similarly metropolitan, but have very different out-of-district contribution rates to electoral candidates[15].

3.2.3 Characterizing Inter-District Funding Flows

Finally, we briefly describe some characteristics of the localized funding network formed by out-of-district donations. Figure 4 displays the relationship between the amount donated out-of-district and the amount received from out-of-district, for each district. For both entities, the majority of districts fall cleanly into two groups; those that send more than they receive (e.g. Trinity—St. Paul's), and those that receive more than they send (e.g. Scarborough Centre).

Table 4 shows the proportion of contributions that are received by neighbouring districts, and by districts out-of-province, among all out-of-district contributions. These proportions offer a

[14]Since districts vary enormously in size and shape depending on their population density, one might naturally expect that tightly-packed urban districts would have higher out-of-district contribution rates, simply due to the spatial nature of socio-political interest. In Appendix F, we simulate the out-of-district share for each district under this expectation, and find that district-level out-of-district proportions differ much more than what can be explained by the spatial arrangement of districts alone.

[15]District-level differences in candidate-bound out-of-district contribution rates are even more pronounced when election periods are examined individually (Appendix G).

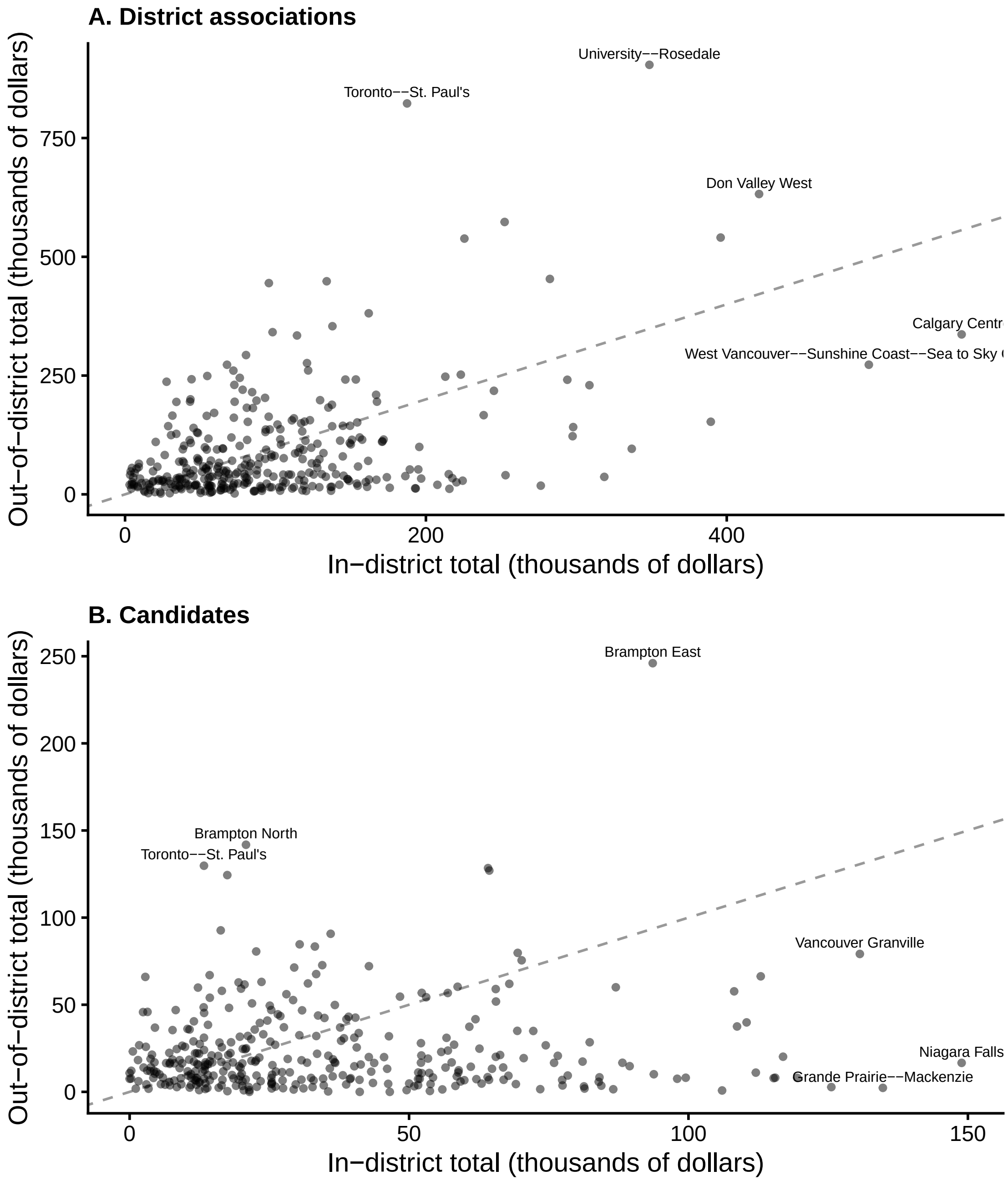


Figure 3: Relationship between the sum of contributions over \$200 donated in- v.s. out-of-district, for each district during the 2013 representational order (effective between 2015 and 2024). (A) shows this relationship for donations to EDAs, and (B) shows this for donations to candidates. Points represent individual districts. The grey dashed line is 1:1. The top 3 districts by amount contributed in- and out-of-district respectively are labelled with their names. Graphs are similar when counts are displayed instead of amounts. (A) is similar when only donations sent during writ periods are considered, and (B) is similar when specific election cycles are considered.

Table 4: Proportion of out-of-district contributions over \$200 that were sent to adjacent districts, and out-of-province during our study period. Raw and value-weighted proportions are reported, along with the overall number of contributions for reference. Contributions received by candidates outside general election cycles (by-elections) are not included in calculations.

Received by	Sent to	By count	By amount
Candidate	Adjacent district	43.3%	44.1%
Candidate	Out-of-province	10.6%	9.7%
District association	Adjacent district	37.1%	36.0%
District association	Out-of-province	14.7%	14.4%

rough picture of how far the average out-of-district donation travels[16]. For both entities, around 40% of out-of-district donations are sent to adjacent districts, while only around 12% reach other provinces. Out-of-district donations are overall closer to home for candidate-bound donations, echoing our findings in Table 2. Those out-of-province donations are visualized further in Figure 5. For donations to district associations, the total flux (sum of contributed amount received and sent) of a province strongly correlates with its population, and contribution rates are larger between two highly populated provinces. The network of interprovincial funding of candidates, however, has different primary edges, and overall contributions are more concentrated between B.C., Alberta, and Ontario. While we should expect reduced interprovincial contribution rates to candidates compared to EDAs (Table 4), certain differences in the candidate network such as Ontario and B.C. concentrating their outflow to Alberta, represent genuine differences in interprovincial funding regimes between these entities.

3.3 Out-of-District-Propensity Models

3.3.1 Individual-Level Model

We used a logistic regression model to explore how the likelihood of out-of-district donation varies across different transactional characteristics, donor demographic environments, and district competitiveness, for individual donations. Our outcome variable is binary: whether or not a donation was sent out-of-district. This model was fit on both donations to EDAs and candidates combined. Models fit on each localized entity separately had similar coefficient magnitudes, signs and significances (see Appendix E). Donations were not grouped by individual,

[16] A distribution of donation distances (measured from the centroid of donor's dissemination area to the centroid of the recipient's district) is displayed in Appendix F.

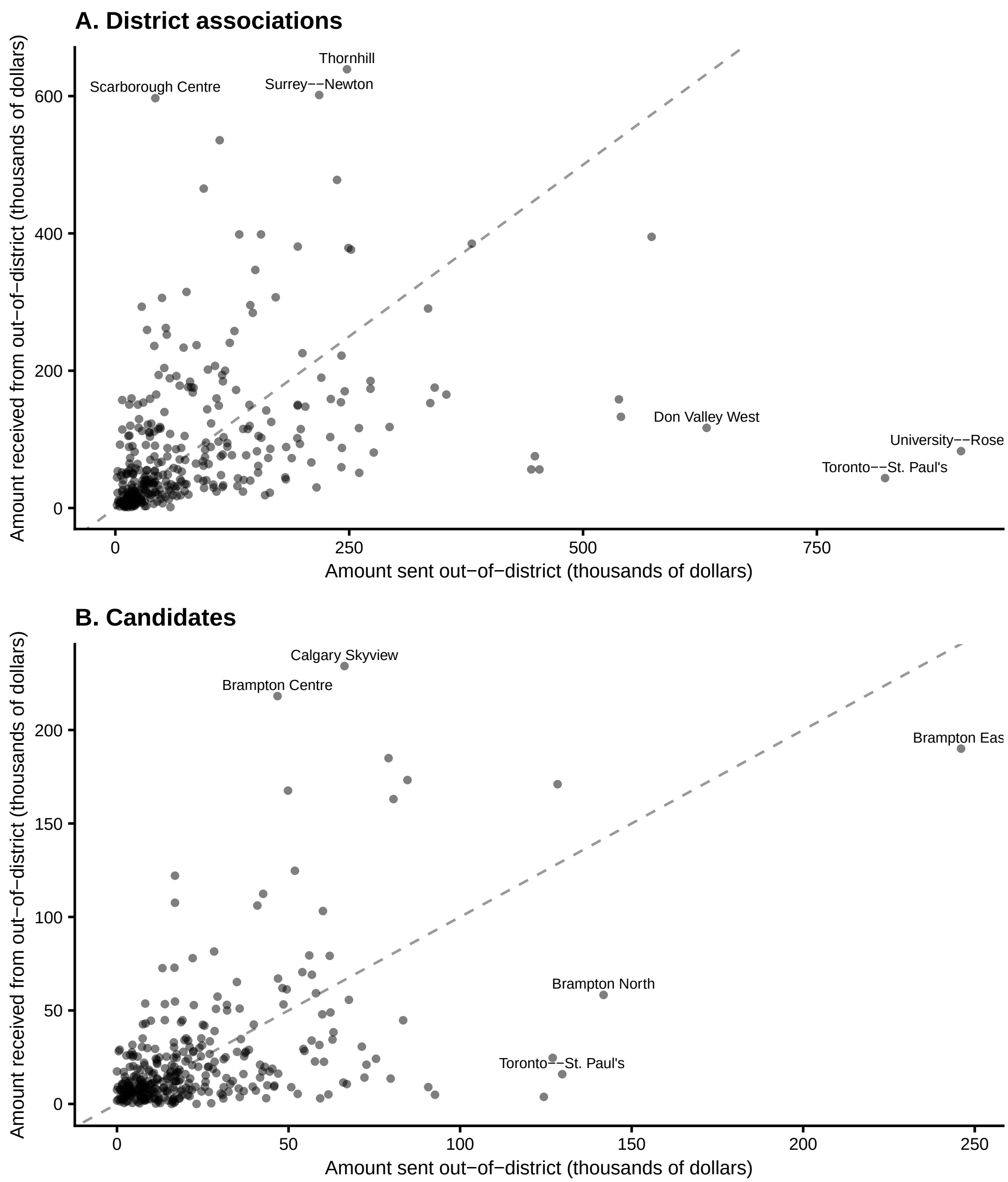
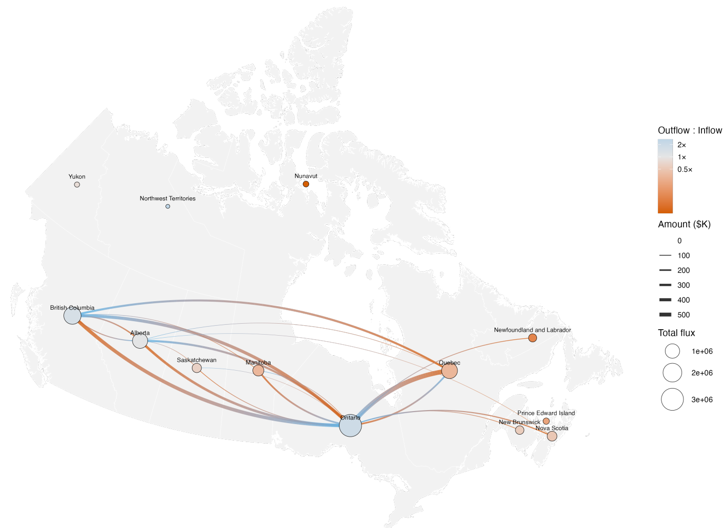


Figure 4: Relationship between the sum of contributions over $200 sent out-of-district v.s. received from out-of-district, for each district during the 2013 representational order (effective between 2015 and 2024). (A) shows this relationship for donations to EDAs, and (B) shows this for donations to candidates. Points represent individual districts. The grey dashed line is 1:1. The top three districts by amount sent and received respectively are labelled with their names. Graphs are similar when counts are displayed instead of amounts. (A) is similar when only donations sent during writ periods are considered, and (B) is similar when specific election cycles are considered.

A. District associations



B. Candidates

Figure 5: Interprovincial funding flows for contributions over $200 during the 2013 representational order (effective between 2015 and 2024). (A) shows flows for donations to EDAs, and (B) shows flows for donations to candidates. Networks were similar when counts were displayed instead of amounts. Node size reflects total interprovincial flux (amount sent + amount received), and edges are weighted by transfer amount. Provinces with red-coloured nodes receive more contributions than they send, and vice-versa for blue nodes. Edges are coloured on a gradient from blue (sending province) to red (receiving province) to mark direction. Candidate networks for individual writ-periods were similar to their aggregate displayed here.

since we were interested in both donor and recipient characteristics in this model. Diagnostics showed that residual autocorrelation, the primary statistical risk associated with this decision, was not a large concern in the model.

The transactional predictors in our model are: the type of recipient (candidate or district association), their political party, and the monetary value of the donation. We did not include time-indexed variables since our data summaries showed that out-of-district rates were constant over time, for both candidates and district associations. Following Garnett (2024), we also considered the following characteristics of the donor's socio-demographic environment: province (grouped into regions), mean age, median household income, the share of the population with a post-secondary degree, and the share of the population that is a visible minority[17]. These predictors are associated with out-of-district donation in the U.S. (Gimpel et al. 2006, 2008). These environmental characteristics were assigned to donations based on their donor's dissemination area. District competitiveness was proxied using the difference between the percentage of votes recieved by the winning and runner-up candidates, averaged across all three general elections captured in our study period[18], inspired by Garnett (2024). This predictor was included since we might expect donors from less competitive districts to donate out-of-district more often, recognizing that a local contribution will do little to affect an outcome that is effectively predetermined.

We also included the population density of the donor's district, for two reasons. First, out-of-district donors are more likely to come from urban (and therefore more densely populated) districts in the U.S. (Gimpel et al. 2008). Second, population density serves as a relatively good proxy for how spatially clustered a district is, since Canadian districts are designed to have similar population sizes (Elections Canada 2025b). Out-of-district donation rates are likely naturally higher in smaller districts, since donors are more likely to interact with individuals, businesses, or politicians close by that happen to be located in a different district. Population density was transformed in the model by applying a square root, allowing us to interpret the metric as the inverse of the average distance between people, which captures this distance effect more faithfully[19].

[17]An alternative model where regions were disaggregated into provinces showed qualitatively similar results. An alternative model where the visible minority proportion was disaggregated into specific ethnic groups is shown in Appendix E.

[18]Districts were more likely to have elected an MP from the same party (or simply the same MP) for all three elections when their average margin of victory is higher (pseudo$R^2 = 0.22$), meaning that this competitiveness metric also captures a decent share of the variation in district-level party stability. These data were sourced from Open Government Canada (Treasury Board of Canada Secretariat 2026).

[19]The observed spatial variation in out-of-district propensity cannot be fully explained by the spatial arrangement of districts alone (Appendix F), however it remains an important consideration.

The resulting model is:

$$\begin{aligned}\text{logit}(p_i) = \beta_0 + \beta_1\text{Election}_i + \beta_2\text{Candidate}_i + \sum_{j=3}^{5}\beta_j\text{Party}_{ji} + \beta_6\text{Amount}_i \\ + \sum_{k=7}^{11}\beta_k\text{Region}_{ki} + \beta_{12}\text{Age}_i + \beta_{13}\text{Income}_i + \beta_{14}\text{Education}_i + \beta_{15}\text{VisMin}_i \\ + \beta_{16}\text{VictoryMargin}_i + \beta_{17}\text{Density}_i\end{aligned}$$

where:

- p_i is the probability that donation i was an out-of-district donation.
- Election_i indicates whether donation i occurred during a general election period.
- Candidate_i indicates whether donation i was addressed to a Candidate (baseline is EDA).
- Party_{ji} indicates which party (among Liberal, N.D.P, or Other, with baseline being Conservative) the recipient is associated with.
- Amount_i is the monetary value of donation i.
- Region_{ki} indicates which region (among Atlantic Canada, B.C., Prairies, Quebec, Territories, with baseline being Ontario) the donor is located.
- Age_i is the average age of residents in the donor's dissemination area.
- Income_i is the median household income in the donor's dissemination area.
- Education_i is the proportion of residents in the donor's home district holding a post-secondary qualification.
- VisMin_i is the proportion of residents in the donor's home district who identify as a visible minority.
- VictoryMargin_i is the difference between the percentage of votes recieved by the winning and runner-up candidates, averaged across all three general elections in the donor's home district.
- Density_i is the population density (persons per km^2) of the donor's home district. As mentioned in-text, the model was fit using the square root of this variable.

3.3.2 District-Level Model

Inspired by the large variation in out-of-district giving between districts observed in Figure 3, we fit a second regression model to explore which socio-demographic features were associated with higher rates of out-of-district donation, aggregated at the district level. Our outcome variable in this case was the proportion of the total contributed amount[20] sent out-of-district during our entire study period, for each district. Given that this outcome is a proportion, we used beta regression, fit using the `betareg` package in R (Cribari-Neto and Zeileis 2010). This model was fit on both donations to EDAs and candidates combined, following from the individual model. A model fit on only EDAs separately had similar coefficient magnitudes, and the same signs and significances (shown in Appendix – E). A model fit on only candidates was not fit due to data sparsity and the presence of singularities in the response.

The covariates included in this model were the same environment variables chosen for the individual-level model, aggregated at the district level[21]. We also included the same district-level competitiveness metric used in the individual model. Since districts had very different numbers of donations used to calculate our outcome proportion, we also assumed a power-law relationship between the common precision parameter ϕ and the number of donations, following Simas et al. (2010).

The resulting model is:

$$\begin{aligned} p_i &\sim \text{Beta}(\mu_i, \phi_i) \\ \text{logit}(\mu_i) &= \beta_0 + \beta_1 \text{Amount}_i + \sum_{j=2}^{6} \beta_j \text{Region}_{ji} + \beta_7 \text{Age}_i + \beta_8 \text{Income}_i \\ &\quad + \beta_9 \text{Education}_i + \beta_{10} \text{VisMin}_i + \beta_{11} \text{VictoryMargin}_i + \beta_{12} \text{Density}_i \\ \log(\phi_i) &= \gamma_0 + \gamma_1 \log(n_i) \end{aligned}$$

where:

- p_i is the proportion of the total contributed amount that was sent out-of-district by donors in district i.
- μ_i and ϕ_i are the mean and common precision parameters of the distribution of p_i.
- n_i is the number of donations originating from district i.
- Amount_i is the total amount contributed by donors in district i.
- Region_{ji} are indicators for the donor region of district i.
- Age_i is the mean age of residents in district i.

[20] We observed high correlation ($\rho = 0.978$) between the proportion of out-of-district donations and the share of total contributed amount that left the district, for both candidates and EDAs. Therefore, results would have been similar if we had used the number of donations as our outcome instead of the amount-weighted share of donations.

[21] As with the individual-level model, alternative model specifications with where disaggregated region and visible minority proportion is shown in Appendix E.

- Income_i is the median household income in district i.
- Education_i is the proportion of residents in district i holding a post-secondary qualification.
- VisMin_i is the proportion of residents in district i who identify as a visible minority.
- VictoryMargin_i is the difference between the percentage of votes recieved by the winning and runner-up candidates, averaged across all three general elections in district i.
- Density_i is the population density (persons per km^2) of district i. As mentioned in-text, the model was fit using the square root of this variable.

4 Results

Here we present the results from our out-of-district propensity models. We note that many of our key findings are best understood as data summaries and visualizations, which are presented in Section 3.2. Section 3.2.1 provides context for how the localized contributions considered in our study fit into the broader scope of financial contributions in general. Section 3.2.2 provides spatio-temporal summaries of out-of-district propensity (which were used to inform the model discussed here). Section 3.2.3 briefly characterizes the funding network formed by out-of-district donations. Both summary-based and model-based findings are discussed together in Chapter 5.

Regression results from the individual- and district-level models are presented in Table 5 and Table 6 respectively. Figure 6 and Figure 7 provide a visualization of coefficient signs, magnitudes, and significances. For both models, exponentiating the coefficients yields a more interpretable value, the odds ratio. For the individual (i.e. logistic) regression model, an odds ratio above one for a categorical variable (e.g. recipient belonging to the Liberal Party) indicates that, holding all else constant, donations of that category have increased odds of being out-of-district, relative to the baseline category (e.g. recipient belonging to the Conservative party). For the district level (i.e. beta) regression model, an odds ratio above one for a categorical variable (e.g. district situated in the Prairies) indicates that, holding all else constant, the expected ratio of out- vs. in-district contributions increases. Odds ratios less than one and continuous variables (e.g. median household income) are interpreted accordingly.

Individual donations are less likely to reach out-of-district when they are bound for candidates instead of district associations (EDAs), and more likely for Liberal recipients instead of Conservative recipients. The contribution likelihoods of Conservative and N.D.P recipients were not meaningfully different. Larger donations were more likely to leave the district compared to smaller ones, helping to explain why amount-weighted proportions in our data summaries were consistently larger. The average margin of victory (taken to be a measure of district non-competitiveness) was, expectedly, positively assocaited with out-of-district propensity. Donations originating from all regions were less likely to reach out-of-district compared to Ontario, with the exception of Quebec, whose donors have nearly 1.5 times the odds to donate out-of-district, holding all else constant. Donations originating in the Territories were especially unlikely to travel outside their district, with nearly one-eighth the baseline Ontario odds. Turning to the donor's socio-demographic environment, regions with higher population densities, post-secondary education rates, median household incomes, and visible minority proportions[22] all favoured out-of-district donation. However, the average age did not significantly impact out-of-district chances in the presence of other socio-demographic variables[23]. When ethnicities were disaggregated, the positive effect of the visible minority proportion was observed for all minority ethnic groups recorded in the Census (see Appendix E).

At the district level, we modelled the ratio of out- to in-district contribution sums. Similarly to the individual models, districts with higher population densities, post-secondary education rates, median household incomes, and visible minority proportions were more likely to allocate more of their contributions to political entities in other districts rather than their own. The average age of a district was, similarly, insignificant in the presence of other socio-demographic variables. Quebec and the Territories were the only regions to differ significantly from Ontario in their out-of-district ratios, and with the same coefficient signs as in the individual model. Unlike the individual model, the total amount of donations raised and the average margin of victory in a district did not have a significant effect on the proportion of that amount that left said district; while donations are more likely to travel out-of-district when they are larger, and when districts are less competitive, these effects seems to be a purely individualized.

The eight variables used in our district-level model were able to explain a large amount of the variance in the ratio of out- to in-district donation ($pseudoR^2 = 0.67$). The same model fit on only population density also achieved a relatively good fit ($pseudoR^2 = 0.45$), however other variables, and especially socio-demographic ones such as educational attainment and visible minority proportion, improved the model's explanatory power well past what population density alone could offer.

[22] This effect was observed even when donations from the four Brampton constituencies were removed, since data summaries showed that these clearly had very high influence on the model fit. A similar check was performed on the district-level model.

[23] Average age was significant in individual and district-level models where it, along with population density, were the only socio-demographic predictors. This tells us that its effect is mostly explained by other factors, such as income or educational attainment.

Table 5: Results from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. The reference entity is the district association, the reference party is the Conservative Party, and the reference region is Ontario. Estimates are bolded when significant at the alpha = 0.05 level. Small losses (~ 1.1% of entries) were incurred due to missingness in the census data.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−2.889**	**0.062**	**0.056**	**0.003**
District Associations	-		-	
Candidates	**−0.150**	**0.015**	**0.861**	**0.013**
Conservative	-		-	
Liberal	**0.319**	**0.014**	**1.376**	**0.020**
N.D.P	0.013	0.018	1.013	0.019
Other	**0.349**	**0.028**	**1.418**	**0.040**
Amount	**0.000**	**0.000**	**1.000**	**0.000**
Ontario	-		-	
Quebec	**0.318**	**0.020**	**1.375**	**0.028**
British Columbia	**−0.431**	**0.018**	**0.650**	**0.011**
Prairies	**−0.386**	**0.019**	**0.680**	**0.013**
Atlantic Canada	**−0.262**	**0.027**	**0.769**	**0.021**
Territories	**−1.995**	**0.145**	**0.136**	**0.020**
Age	−0.002	0.001	0.998	0.001
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.923**	**0.059**	**6.841**	**0.406**
Visible Minority	**2.205**	**0.029**	**9.071**	**0.260**
Margin of Victory (mean)	**0.007**	**0.001**	**1.007**	**0.001**
Population Density (sqrt)	**0.003**	**0.000**	**1.003**	**0.000**
Num.Obs.	136 437		136 437	
AIC	164 482.3		164 482.3	
BIC	164 649.3		164 649.3	
Log.Lik.	−82 224.138		−82 224.138	
F	1115.752		1115.752	
RMSE	0.46		0.46	

Table 6: Results from a beta regression model where the outcome is the out-of-district share of the sum of contributions over $200 originating from an individual district during our study period. The reference entity is the district association, and the reference region is Ontario. Estimates are bolded when significant at the alpha = 0.05 level. Small losses (~ 1.1% of individual transactions) were incurred due to missingness in the census data. Each of the 338 districts was represented in this model.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−4.276**	**0.797**	**0.014**	**0.011**
Amount	0.000	0.000	1.000	0.000
Ontario	-		-	
Quebec	0.556	0.112	1.744	0.196
British Columbia	−0.166	0.098	0.847	0.083
Prairies	−0.144	0.109	0.866	0.094
Atlantic Canada	**0.228**	**0.138**	**1.256**	**0.173**
Territories	**−1.339**	**0.483**	**0.262**	**0.126**
Age	0.015	0.016	1.015	0.016
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.654**	**0.600**	**5.226**	**3.137**
Visible Minority	**1.918**	**0.221**	**6.805**	**1.505**
Margin of Victory (mean)	0.004	0.003	1.004	0.003
Population Density (sqrt)	**0.012**	**0.002**	**1.012**	**0.002**
Num.Obs.	338		338	
R2	0.672		0.672	
AIC	−473.8		−473.8	
BIC	−416.4		−416.4	
RMSE	0.13		0.13	

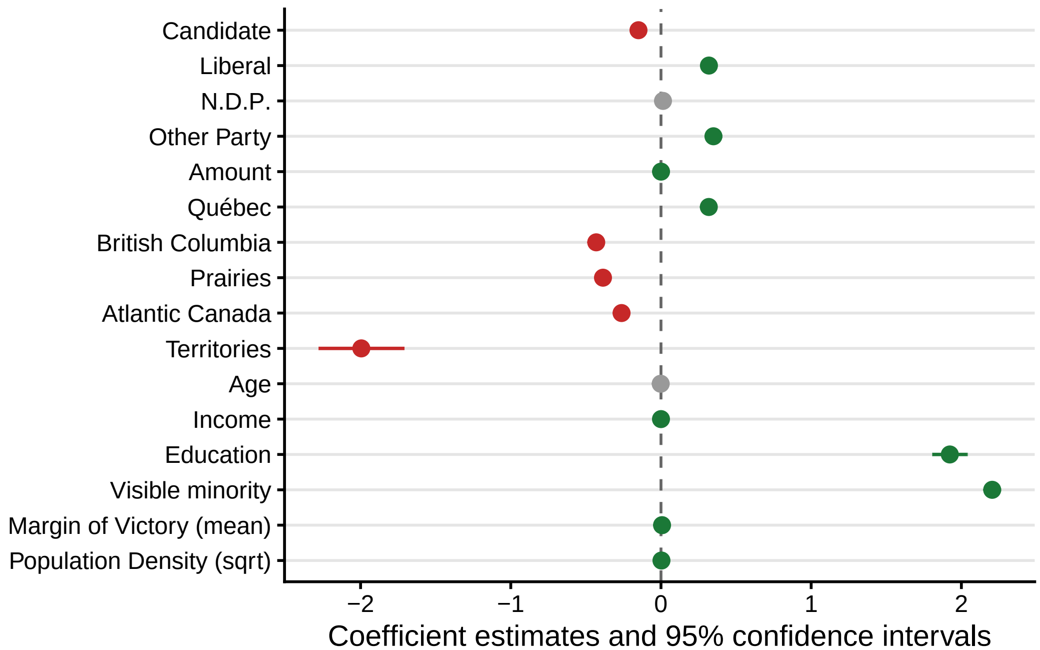


Figure 6: Coefficient estimates and 95% confidence intervals from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. Coefficients are coloured grey if they are insignificant at alpha = 0.05, red if they are significant with a negative effect, and green if they are significant with a positive effect. Reference entity is the district association, reference party is the Conservative Party, reference region is Ontario.

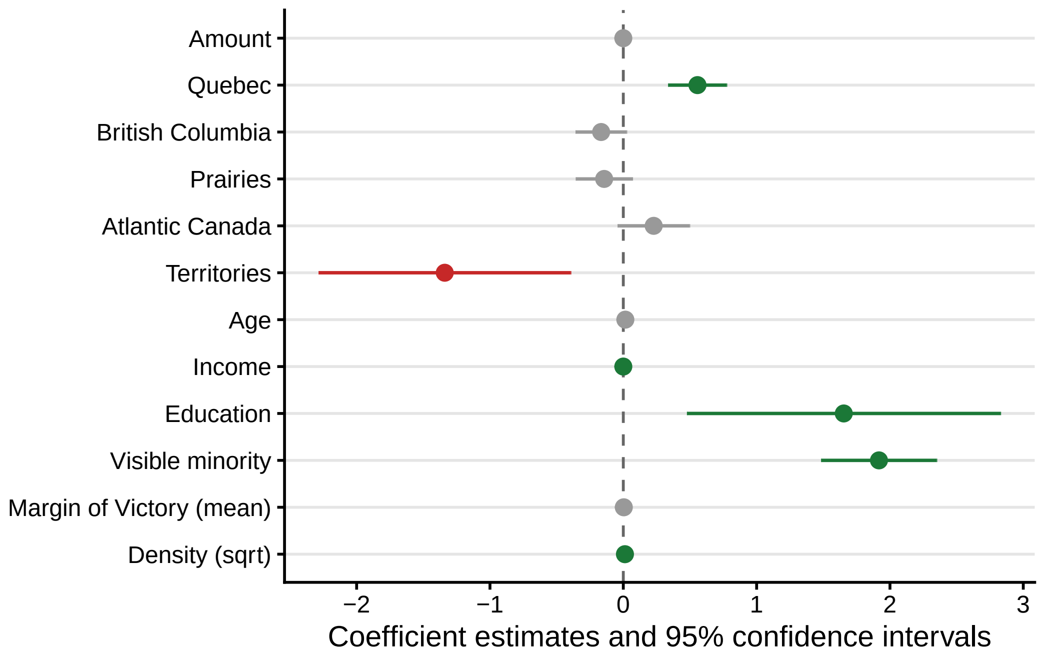


Figure 7: Coefficient estimates and 95% confidence intervals from a beta regression model where the outcome is the out-of-district share of the sum of contributions over $200 originating from an individual district during our study period. Coefficients are coloured grey if they are insignificant at alpha = 0.05, red if they are significant with a negative effect, and green if they are significant with a positive effect. Reference entity is the district association, reference region is Ontario.

5 Discussion

In this paper, we used publicly available political financing data to document the extent and spatio-temporal trends in Canadian out-of-district donations from 2015 to 2024. Further, we identified donor-environmental characteristics associated with out-of-district propensity (i.e.: where out-of-district donations are more or less likely), and provided a preliminary overview of the spatial network formed by out-of-district donations. Our findings are also contextualized within the broader scope of all[24] Canadian federal political contributions over $200 during this time (see Section 3.2.1). We present and discuss our major findings here, along with some notable study limitations.

Out-of-district contributions constituted a substantial source of funding during our study period, to electoral campaigns and district associations alike. Naturally, overall contribution rates to these two political entities varied in time; campaign funding is largely restricted to writ periods, and donations to district associations swelled during the holiday season and electoral cycles, which garnered nearly five times more contributions over $200 compared to baseline rates. However, for both entities, the pooled share of funding received from outside their district (equivalently, the share of the total contributed amount that left its donor's district) remained largely constant throughout the decade. Roughly 40% of individual donations over $200 (accounting for 44% of the total amount contributed) are sent out-of-district, a rate which remained stable throughout our study period. This rate and its consistency through time contrasts against U.S. findings, where the bulk of existing out-of-district donations research has been done. There, out-of-district funding shares to senators and congressional representatives during electoral campaigns have increased since the start of the 21st century (Gimpel et al. 2006; Canes-Wrone and Miller 2022; Sievert and Mathiasen 2023), and the average representative in 2016 received around two-thirds of their contributions from out-of-district (Canes-Wrone and Miller 2022).

While out-of-district contribution rates varied little over time, across parties, and between the recipient entity types that we considered (candidates or district associations), we document drastic differences in out-of-district propensity from district-to-district over our study period, both at the level of individual donations, and entire districts. For example, out-of-district donation shares to district associations over the entire studied decade were as little as 5% for some districts (e.g. Niagara Falls), and as high as 75% in others (e.g. Trinity — St. Paul's). While we might expect passively higher out-of-district rates solely due to the spatial proximity of other districts, empirical variation in rates of out-of-district giving (aggregated at the district

[24] Excluding a small number of contributions (1.2%) where we were not able to geolocate the donor.

level) far exceeded simulated expectations under this hypothesis. Further, most deviations from expected out-of-district donation rates were districts with higher, not lower observed out-of-district contribution shares (see Appendix F). This variation is also not simply a result of the large differences in the number of donations receive by different districts; restricting our sample to only those districts among the top 10% by total contributed amount yielded a similarly broad range of out-of-district propensities.

Donations originating from densely populated, high-income, and well-educated districts are more likely to travel out-of-district, all else held constant. This set of covariates agrees with findings established in the U.S. (Gimpel et al. 2008). These are also significant predictors of donation rates in general, across North America (Gimpel et al. 2008; Garnett 2024). In addition to these, however, the visible minority proportion of a donor's home environment is positively associated with out-of-district giving, and was the single most explanatory socio-demographic predictor in both of our models. This finding departs from the established profile of American "donor districts", which historically have predominantly European ethnic makeups (Gimpel et al. 2008). However, recent Canadian studies support this inverted relationship between ethnicity and political engagement. Garnett et al. (2022), among others, note that some minority populations such as South Asians donate at rates that far exceed their share of the population, and, topically, Besco and Tolley (2022) found that individual donors belonging to most minority ethnic groups show higher rates of out-of-district donation compared to donors of European descent. The strong relationship between highly minoritized regions of the country and rates of out-of-district giving identified in our study is most plausibly attributed to one of two mechanisms which our models, as specified, cannot disentangle; demographic collinearity, and individual co-ethnic affinity. First, the relationship may simply be a byproduct of Canada's unique social demography. Visible minority populations in Canada are concentrated almost exclusively in cities, and higher minority composition is not obviously associated with lower income (a well-established predictor of donation and out-of-district donation propensity) at the national level (Hou 2004; Walks and Bourne 2006). In the U.S., however, areas with high visible minority proportions do not follow an urban-rural gradient nearly as closely (Lee and Sharp 2017) and do have lower income, on average (Quillian 2012). Therefore, it is entirely possible that our visible minority coefficient is absorbing residual features of affluent, dense, urban Canada that density and income alone do not fully capture, rather than reflecting anything about ethnicity itself. More generally, collinearity among socio-demographic predictors was present in our dataset, making it difficult to disentangle the specific drivers of out-of-district donation, even though it is undeniable that social demography and out-of-district donation are linked, beyond simple population density. Alternatively, this relationship may be signalling the presence of co-ethnic inter-district donor networks. Identity-based donor affinity is a well-established feature of Canadian political participation (Besco 2019; Besco and Tolley 2022;

Tolley et al. 2022). Further, the largest component of our inter-district donation network we could identify is the flow of funds between the four districts that make up the Brampton region within the Greater Toronto Area, whose population is majority South Asian. While attractive, this hypothesis cannot be confirmed without individualized ethnic identification data, largely inaccessible at this scale.

Despite empirically high rates of out-of-district donation, our study provides evidence that these donations remain a largely regionalized phenomenon. Nearly half of donations that leave their district during our study period were destined for entities in adjacent ones, and only around 12% ever even left their donor's province (specifically, 9% for candidate-bound donations). U.S. research on out-of-state donations describes highly nationalized donor bases among incumbent and campaigning House representatives (Gimpel et al. 2006; Canes-Wrone and Miller 2022; Sievert and Mathiasen 2023; Keena 2024). For example, a study on senatorial elections from 2000 to 2020 found that both Republican and Democrat senators received at least 49% of their donations from out-of-state during election seasons (Sievert and Mathiasen 2023). Some of this difference may simply be a matter of spatial proximity; U.S. states are closer to each other than Canadian provinces and territories. The sheer scale of this difference, however, speaks to a more fundamental difference between the geographical donor bases of each country; Canadian electoral candidates largely source individual-based funding from within their own province. For those few donations bound for other provinces, we observe qualitatively different patterns for candidate-bound and district association-bound donations. Further research on Canadian out-of-district donations may take advantage of the spatial network they form to explore these patterns further.

Our findings are accompanied by some important limitations. First, a non-negligible proportion (34%) of the full amount addressed to localized entities were under $200, and therefore not included in our study. The majority (75%) of these donations were location-anonymized, and were excluded out of necessity, following Gimpel et al. (2008) and Garnett (2024). Distributional comparisons between study and location-anonymized contributions (see Appendix D) showed that this missing sum was distributed similarly to the study data across recipient characteristics such as party affiliation, region, and over time, suggesting that their exclusion introduced little systematic bias into our study dataset. However, location-disclosed contributions under $200 were distributed very differently to those above $200 included in our study: to an overwhelming degree, these were in-district donations to Liberal district associations outside of election cycles, and were considerably less likely to be sent out-of-district than those in our study (18% compared to 42%). Further investigation into these cases revealed that only 12.4% of these transactions had a unique donor name, postal code, and year, compared to 92.7% uniqueness across the same set of factors in our study data. This tells us that a majority of these transactions are likely from donors who, in a year, contributed a total above $200 to district associations, and therefore had their locations disclosed. Given this, the exclusion of location-disclosed contributions under $200 likely inflates our estimates of out-of-district rates for Liberal district associations, which were already found to receive the largest contribution share from out-of-district among all entity-party pairs (57% compared to the average of 40%). More broadly, this finding, paired with the positive association between amount donated and out-of-district propensity, suggests that our estimates of out-of-district donation rates, particularly the amount-unweighted point estimates, are overestimates of the true underlying out-of-district propensity. Nevertheless, since contributions under $200 remained similarly distributed to the study data, particularly in space, their exclusion is unlikely to have greatly affected the statistical significance of our most interesting covariates in our out-of-district propensity models, including those related to social demography.

Second, our focus on out-of-district contributions meant excluding donations addressed to national entities (parties and leadership contestants). As shown in Section 3.2, contributions of this type represented a majority (around 80%) of contributions during our study period. The exclusion of these entities therefore limits the generalizability of our findings; the majority of Canadian donation cannot actually be assessed as having been sent in- or out-of-district. While federal political donation in general is best understood as a nationalized phenomenon, our study provides evidence that the individual donor bases for two key players on the Canadian federal political financing scene – district associations and electoral candidates – are largely regionalized. District associations and candidates receive some resources through internal party transfers; however, these entities continue to greatly depend on individual contributions, making donor geography an important dimension of their financial support base. Further research may use our dataset of donor-geolocated political contributions to investigate whether national party donor bases are similarly constrained to particular regions of the country.

Third, our chosen study period (October 2015 to April 2024) truncates the available data in the middle of the 44th parliament. This was done to restrict our analysis to a single, static set of districts. While we succeeded in excluding non-2013-order districts, we cannot guarantee that all valid 2013-order FED entries were captured[25], since the adoption of new districts is a multi-year undertaking, and timelines can differ between locations (Elections Canada 2025b). Further, our chosen date range results in the third non-election period of our analysis being incomplete, so we cannot claim to have studied three full parliaments. However, we did not identify any systematic differences in out-of-district donation rates between parliaments, having

[25] Over all entries in the dataset which had recipients belonging to 2013-order districts (2004 - present), only 0.5% were not captured by our chosen range.

mainly discovered variation through space, and not through time.

Appendix

A Background on Canadian Political Financing

A.1 Individual Contribution Limits

The *Canada Elections Act* imposes individual contribution limits on donors seeking to contribute to one of the following registered, federal political entities[26]: political parties, electoral district associations (EDAs), candidates, party nomination contestants, and party leadership contestants (Elections Canada 2026a). These limits are shown in Table A1[27] and Table A2. On top of these, candidates are also permitted to give an additional $1,775 in total per year in contributions to other candidates, registered associations, or nomination contests, including their own. Regulated expenses for any of the entities themselves cannot exceed specific values, which are determined on an annual basis, and are different for each riding (see Elections Canada 2026b). Individual contributions to registered third-party organizations are not regulated by the *Act*, and were not included in this study (Elections Canada 2026f).

Only Canadian citizens or permanent residents can make contributions (Elections Canada 2026f). Any donor who contributed $20 or more to any of the five federal political entities listed above must have their name recorded (but not reported to Elections Canada). Donors with contributions exceeding $200 must also have their names and addresses recorded and reported to Elections Canada. Cash donations above $20 are prohibited (Elections Canada 2026f).

A.2 Financial Reporting Requirements of Federal Political Entities

All of the information presented in this section can be found on the "Political Financing" page of the Elections Canada website (Elections Canada 2026e). Financial returns represent an account of an entity's inflow (i.e. contributions, loans or transfers) and outflow (expenses). At minimum, all entities must submit financial returns annually (for parties and EDAs), or within 4 to 8 months after an electoral event (for parties, candidates, third parties, leadership contestants, and nomination contestants exceeding $1000 in contributions or expenses). Returns with large transaction amounts must be accompanied by an external audit. Entities such as parties and EDAs may be required to file reports on a quarterly basis. Some (but not all) returns are

[26] A full definition for each entity is available at Elections Canada (2026d).
[27] Contribution limits presented here are only for 2026. They increase by $25 every year on January 1st.

Table A1: 2026 Annual Limits on Individual Contributions, Loans, and Loan Guarantees (adapted from Elections Canada website).

Political Entity	2026 Annual Limit (CAD)
To each registered national party	$1,775
In total, to all leadership contestants in a particular contest	$1,775
In total, to all EDAs, nomination contestants, and candidates (for each registered party)	$1,775
In total, to each independent or non-affiliated candidate	$1,775

Table A2: 2026 Limits on Individual Contributions, Loans, and Loan Guarantees, for Politicians Contributing to Their Own Campaigns.

Politician	Own-Campaign Contribution Limit (CAD)
Nomination Contestant	$1,000 (in addition to regular annual limits)
Candidate	$5,000
Leadership Contestant	$25,000

reviewed by Elections Canada auditors. Risk-based auditing is used to determine which files will be audited and what level of audit will be performed (Elections Canada 2026c).

A.3 Electoral Redistricting

Electoral district boundaries are subject to change every 10 years, as required by the *Constitution of Canada* (Elections Canada 2025b). The 2013 representational order consisted of 338 districts, and the 2023 representational order consisted of 343 districts. The boundaries of existing districts are subject to change as well. The 2013 representational order was decidedly in effect between the calling of the 42nd general election on October 2, 2015, and the first dissolution of Parliament occurring after April 22, 2024 (Elections Canada 2023).

B Donations Data Pipeline

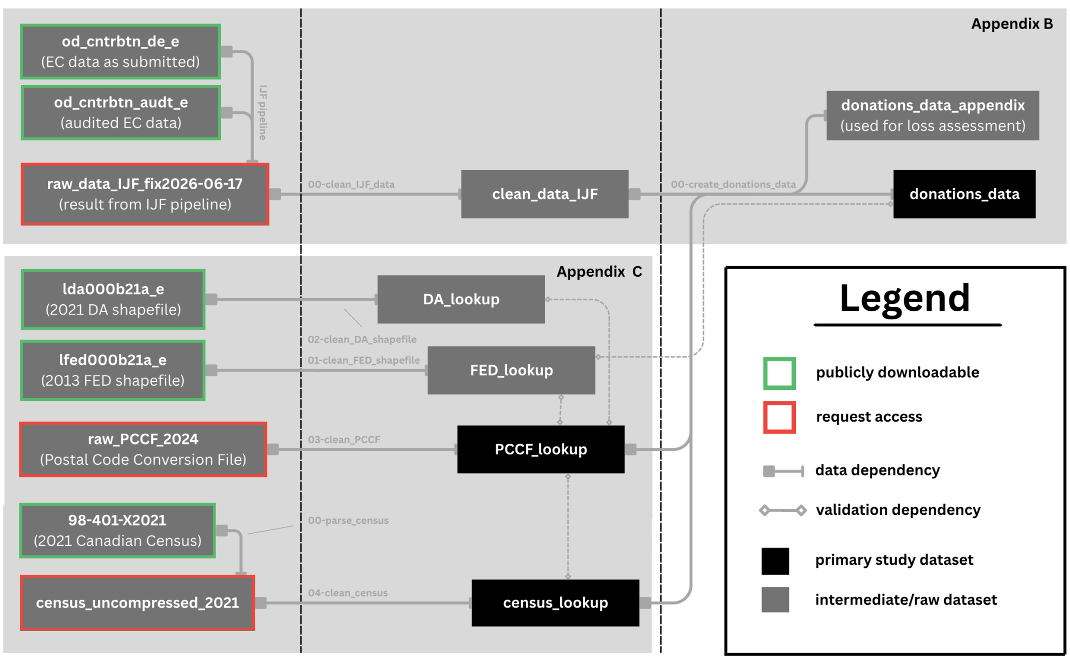


Figure B1: Flow diagram illustrating the full data pipeline used throughout this analysis. All scripts referenced in this figure can be accessed at https://github.com/bennypk1/geolocating-political-donations. This diagram was made using the Canva graphic design software.

In this section we describe, in-detail, the three major cleaning and validation scripts used to transform contributions data from Elections Canada into the dataset used throughout our analysis. These scripts rely on other government datasets, cleaned separately, documented in Appendix C. The full pipeline, including the steps described in Appendix C, is illustrated in Figure B1.

B.1 Investigative Journalism Foundation Pre-Cleaning

The original contribution datasets used in this study can be found on Elections Canada's "Open data on contributions" page, under the heading "Data sets updated" (Elections Canada 2025a). Two products are offered. The first is a dataset of contribution reports filed by all Canadian registered federal political entities (see list of these in Appendix A.1) from January 2004 to present (Elections Canada 2025a). The second is an audited version of the first. Due to pre-screening policies and administrative lag, only some of the first dataset is audited. In our case, the audited version of the dataset has only 61.7% as many entries as the full un-audited data (n = 10095198 ; both accessed on May 5, 2026). Both datasets are updated on a weekly basis by Elections Canada (Elections Canada 2025a), as new audits and financial reports are filed.

Staff at the Investigative Journalism Foundation created a data pipeline to aggregate, de-duplicate, and clean these datasets. Both audited and un-audited versions of the same record were available for most records. To avoid duplicating these, audited and un-audited versions of the same record were linked by assuming that rows sharing the same donation date, electoral event, recipient, and recipient political party referred to the same entry. We note that this linkage approach carries the small risk that two distinct donations to the same recipient on the same date are incorrectly treated as duplicates. The pipeline preferentially keeps audited records when they are available.

Further de-duplication was effectuated differently based on the type of contribution being recorded. For loans, sums reported during an Annual filing might appear in a Quarterly report filed in the same year. In such cases, Annual records were preserved and Quarterly records were discarded. For all other types of contributions, only true duplicates (i.e. rows with identical entries identical in every field) were discarded. Additionally, some entries represented aggregated sums of contributions from anonymous donors. For these, Elections Canada reports a cumulative total per recipient Annually and Quarterly, meaning that the same aggregate sum could appear twice in the same year. To avoid double-counting these, only the largest reported aggregate sum per recipient per year was retained. Finally, a few smaller cleaning procedures were applied to the columns themselves; Table B1 shows where each of the columns present in the IJF data were drawn from columns at the source, and any notable procedures applied to them.

Table B1: Description of where the columns in the IJF data were derived from the source, and any cleaning procedures applied. Variables added for internal database consistency (e.g. processing dates) and redundant variables (e.g. donation year) are not shown here.

IJF Column	EC Source Column(s)	Notes
donor_full_name	Contributor name ; Contributor last name ; Contributor first name ; Contributor middle initial	Concatenated source columns, delimited by commas.
donor_type	Contributor type	No changes from source.
donor_location	Contributor City ; Contributor Province ; Contributor Postal code	Concatenated source columns, delimited by commas.
donation_date	Contribution Received date ; Fiscal/Election date	Used 'Contribution Received date' when available, filled with Fiscal/Election date otherwise. Flagged when imputation occured. Removed invalid dates.
electoral_event	Electoral event	Donations to leadership and nomination contestants had NA entries here. Otherwise, no changes from source.
amount_monetary	Monetary amount	No changes from source.
amount_non_monetary	Non-Monetary amount	No changes from source.
recipient	Recipient ; Recipient last name ; Recipient first name ; Recipient middle initial	Concatenated source columns, delimited by commas.
electoral_district	Electoral District	No changes from source.
political_entity	Political Entity	No changes from source.
political_party	Political Party of Recipient	No changes from source.

B.2 Initial Cleaning of IJF Data

We applied further cleaning and filtering steps to the data supplied by the Investigative Journalism Foundation. First, we removed redundant columns (e.g. data had columns `year` and `donation_year` which were identical for every entry) and columns used for internal consistency (e.g. the date the entry was added by the organization). Next, we removed entries with malformatted dates, or disagreements between the year and date columns. Less than 0.001% of entries were discarded at this step, all corresponding to donations dated in later months of 2026. We then filtered for only donations stated to have been made by individuals. Less than 0.4% of the data was removed at this step, the overwhelming majority of which were records dated before 2006, outside the scope of our study. Next, we filtered for donations where the sum of monetary and non-monetary contributions was present and positive. Less than 0.02% of the data was removed at this step. We then removed entries where the donor or recipient name was either missing or shorter than 4 characters (after trimming whitespaces). Less than 0.001% of data was removed at this step, most of which also had missing donor locations. Finally, we removed individual donations with amounts larger than $25000. Here we were careful to exclude entries that represented aggregated sums of anonymous contributions, which were flagged by searching for entries where `donor_full_name` started with "Contribut". Less than 0.001% of the data was removed at this step. In total, 0.4% of entries (n ~ 30,000) were discarded in this section.

Donor postal codes were extracted from the `donor_location` column using regex-based string matching. Canadian postal codes always follow the format A0A0A0, where 0 is a digit and A is a letter (excluding I, O, D, F, Q, or U). Also, they may not start with W or Z (Canada Post 2026). These facts were used to verify the structure of extracted postal codes, and apply the following imputations:

- "O"s in a number slot were taken to mean "0"
- "I"s in a number slot were taken to mean "1"
- "!" in a number slot were taken to mean "1"

After imputations, we were able to recover 99.89% of individual donor postal codes. The remaining 0.11% were kept, with their postal codes set to `NA`. We note that while these postal codes are guaranteed to be structurally correct, we did not validate whether the codes matched their associated provinces, cities, or were actually used. The effective validity of these postal codes is verified in Appendix B.3. 0.35% of entries (n ~ 26,000) represented aggregated sums of anonymous contributions, and therefore did not have associated postal codes. Despite their scarcity, these amounted to 24.1% of the total contributed amount in the dataset. These were kept at this stage.

We also performed simple string cleaning procedures such as whitespace stripping and entity matching (e.g. United Party of Canada (UP) and United Party of Canada clearly refer to the same political party). Notably all donation recipients in the dataset had associated electoral districts (apart from leadership contestants and registered parties, since these are non-local entities) and political parties. Spot checking revealed no district or party attribution errors among a small set of recipients, however we could not completely confirm that recipient districts or parties were correct.

B.3 Clean IJF to `donations_data`

In this section we describe how our primary study dataset was derived from the cleaned IJF data, and augmented to include the donor's district using a custom postal code conversion file (PCCF) described in Appendix C.1. The resulting dataset represents all donor-geolocated contributions above $200 made out to federal political entities (excluding nomination contestants and candidates during by-elections).

First, we restricted the dataset to only those entries with donation dates between October 2, 2015, and April 22, 2024, capturing the effective period of the 2013 representational order (see Appendix A.1). Next, we removed entries where the recipient's district was not one of the 338 possible districts in the 2013 representational order, referencing our modified PCCF (0.002% of cases). Next, we removed entries where the reported electoral event did not happen within our date range (e.g. 39th general election was in 2006). Although it is possible that these were retroactively added for past events, less than 0.001% of the data had this issue, so their removal did not impact our analysis. Next, we removed individual donations addressed to localized entities which had amounts larger than what the contribution limits allowed (0.0001% of cases). For candidates this was $5000, and for registered associations this was $3550. Next, we removed donations addressed to leadership contestants for parties where no such contest occurred during our study period (0.0001% of cases). Nomination contestants were removed entirely, to simplify analysis. 3% of donations (7% of summed total) received during general election periods had receipt dates outside the stated period (usually dated after the writ period had ended, not before it started). These were not removed due to their non-trivial abundance, however many analyses and visualizations split the data into donations received during vs. outside writ periods. In those cases, these data are considered to have been received outside a writ period. All such anomalies originated from financial reports filed by electoral candidates.

As mentioned in the main text, our study considers only donations above $200, since these are guaranteed to have recorded donor postal codes. This step discards 45% of the total contributed amount during our study period, but only 32% of the total amount received by localized entities (i.e. candidates and district associations). These excluded data are discussed further in Appendix D. For the remaining entries, donor postal codes were matched to their corresponding census dissemination areas (DAs) and federal electoral districts (FEDs) using our modified PCCF file. 1.2% of donor postal codes (representing 1.3% of the total amount contributed) were either missing or could not be matched, and were dropped. Rates were 2.1% and 2.4% for localized entities. Figure B2 breaks down the discarded data by its cause, for localized entities (since these are the primary focus of our study). Finally, columns were renamed, merged, and variables were joined to improve the dataset's usability. The resulting schema is shown in part in Table B2. We provide a list of notable operations here below:

- Monetary (99.8%) and non-monetary (0.2%) contributions were joined, and analysis was carried out on the total.
- Specific electoral events were grouped into the categories: Leadership contest, Nomination contest, General election, By-election, and Non-electoral (i.e. annual or quarterly filings).
- We added a variable to indicate during which writ period (if any) a donation was received, based on `donation_date`. Writ periods were taken to be August 2 to October 19 (2015), September 11 to October 21 (2019) and August 15 to September 20 (2021). By-elections are therefore not considered in the election vs. non-election period distinction, however they only account for 0.1% of the total contributed amount during the study period.

C Supplementary Datasets

In this section we describe how we created two custom lookup tables that the cleaning scripts in Appendix B rely on; one to convert between different levels of spatial hierarchy, and a second to retrieve specific socio-demographic statistics at the census dissemination area. Their place in the full pipeline is illustrated in Figure B1.

C.1 Spatial Lookup Table (`PCCF_lookup` on GitHub)

The purpose of the table constructed in this section is to provide unambiguous links between any administrative area of interest and another at a larger scale (e.g. postal code to district, dissemination area to province, etc.). This custom table is a version of Statistics Canada's 2024 postal code conversion files (PCCF), modified to resolve many-to-many spatial overlaps between postal codes, dissemination areas (DAs), and electoral districts (FEDs). The PCCF we used provides linkages between all registered Canadian postal codes (accurate as of June 2024) and various administrative divisions, including the dissemination areas (DAs) of the 2021

Table B2: Primary variables used in our study, with examples and descriptions. Although not shown, additional information used such as census variables or geographic attributes of districts are easily retrived by relating a donor's dissemination area or district to custom-made lookup tables.

Variable	**Example**	**Description**
donor_name	Alexandra Lulka	Donor full name.
donor_dissemination_area	35202387	Dissemination area (UID) associated with donor postal code.
donor_district	Eglinton–Lawrence	Electoral district associated with donor postal code.
donation_date	2015-10-18	Date donation was received.
electoral_event	General Election	Electoral event (if any) occurring during donation.
total_amount	563.3	Summed monetary and non-monetary value of the donation, in CAD.
recipient_name	Joe Oliver	Recipient full name.
recipient_district	Eglinton–Lawrence	District of the recipient (if applicable).
political_entity	Candidates	Entity type of the recipient.
political_party	Conservative Party of Canada	Recipient party affiliation (if any).
is_out_of_district	FALSE	Whether donor and recipient districts are different.

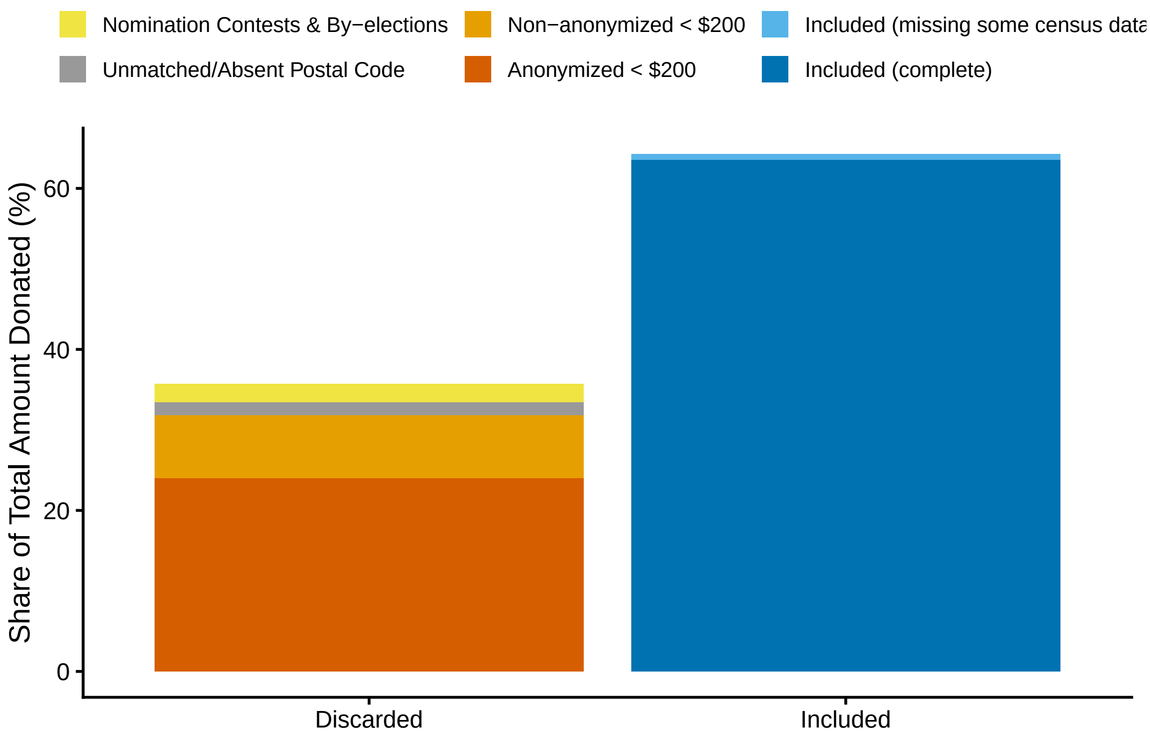


Figure B2: Total amount donated to localized entities (candidates and district associations) during our study period, split by whether or not the share was included in the study. The share of discarded contributions (left) is further broken down by the cause of its removal. We also display the share of included data to which not all census variables could be attributed (light blue).

Census, and the federal electoral districts (FEDs) of the 2013 representational order (Statistics Canada 2025). These sets of DAs and FEDs are the same versions we chose for our study. Access to this PCCF was granted to us by the University of Toronto. The result of this process is shown in part in Table C1.

To resolve cases where one postal code straddled multiple dissemination areas, we used a single link indicator (SLI) to establish a one-to-one relationship between postal codes and DAs, available in PCCF files by default (Statistics Canada 2025). Statistics Canada determines SLIs by estimating the population that resides in each DA within the postal code area (based on 2021 Census population counts) and assigns the code to the one with a higher population (Statistics Canada 2025). This procedure removed 35% of links present in the original PCCF, however no single postal code was removed from the file entirely. A small number of postal codes (0.7%) were removed for missing either an associated DA or FED. All postal codes had associated provinces. In addition, 1.6% of dissemination areas in the dataset straddled multiple district boundaries. To resolve this, for each overlap case, we (1) counted how many postal codes within the dissemination area linked to each candidate FED, and (2) assigned the entire DA, and all of its postal codes to the FED holding the plurality of those postal codes, discarding the postal codes that mapped to any minority FED. Ties between FEDs were broken deterministically by FED UID. This resolution step resulted in the loss of roughly 0.5% of postal codes. Applied to the contributions dataset used in this study, these resolutions increased the share of entries whose donor postal codes could not be matched to the PCCF from 0.7% to 1.2%.

As a final step, we ran a suite of tests, primarily to confirm that the resolved table formed a strictly nested hierarchy: each postal code mapped to a single DA, each DA or postal code to a single FED, and each FED, DA, or postal code to a single province. We also verified that every postal code appeared exactly once, and that they were correctly formatted following Canada Post's guidelines (Canada Post 2026). We also confirmed that all DA and FED UIDs could be cross-referenced with Statistics Canada's 2021 DA and 2013 FED shapefiles (Statistics Canada 2022a), and that all 338 districts, 10 provinces, and 3 territories remained represented. Because of these tests, this spatial lookup table was treated as an authoritative list of valid postal codes, dissemination areas, districts, and provinces, which greatly simplified our analysis.

C.2 Census Lookup Table (`census_lookup` on GitHub)

The purpose of the table constructed in this section is to provide specific socio-demographic variables (listed in Table C2) for every dissemination area (DA) present in the spatial lookup table (described in Section C.1). Socio-demographic information at this spatial scale was

Table C1: Custom spatial lookup table, modified from Statistics Canada's Postal Code Conversion File. This table unambiguously links each Canadian postal code to a 2021 census dissemination area, and each census dissemination area to a federal electoral district from the 2013 representational order. Only a sample of four entries are shown.

Postal Code	Dissemination Area	Federal Electoral District
M4P1E4	35202387	Eglinton–Lawrence
K1A0A6	35061205	Ottawa Centre
V6B1A1	59153201	Vancouver Centre
H3A0G4	24661048	Outremont

sourced from the 2021 Census, and specifically from the "Canada, provinces, territories, census divisions (CDs), census subdivisions (CSDs) and dissemination areas (DAs)" available for download online (Statistics Canada 2022b). We chose the 2021 Census over the 2016 Census since the midpoint of our time period (2019) is closer to 2021 than 2016. We chose the dissemination area, an administrative region that houses roughly 400 to 700 people, since it is the smallest spatial scale where each of our desired statistics are recorded (Statistics Canada 2021). All DAs listed in our spatial lookup file (see Appendix C.1) were present in the Census dataset we used. Since these DAs are the only ones we would ever use for analysis, the other 15.1% of DAs were dropped.

3.2% of remaining DAs were missing at least one census variable: 0.05% were missing all demographic variables, 0.43% only had population, area, and density, 0.12% were only missing the 25% sampled variables (i.e. ethnicity and education level), and 2.62% had all variables recorded except household income. Notably, all entries' missingness could be grouped into five strictly hierarchical stages, summarized in Table C3. Entries with missing variables were not dropped, but were instead flagged so that they could be easily identified during analysis. We also converted ethnicity and education counts to proportions. Table C4 provides a glimpse of the resulting Census dataset.

D Characterizing Excluded Contributions Under $200

In our analysis of out-of-district political donations addressed to electoral candidates and district associations, we excluded donations under $200, and donations whose donors could not successfully be geolocated, which in total represents 33.9% of the total amount received by these entities during our study period (see Table D1). In this section, we compare the recipient characteristics of contributions under $200 with those kept in our study to assess the impact of

Table C2: Census IDs, descriptions, and sample sizes for each census variable used in this study. Variables with a 25% sample size were recorded for only 25% of the population, to preserve privacy.

Characteristic ID	Description	Sample Size
1	Population, 2021	100%
6	Population density (per square kilometres)	100%
7	Land area (square kilometres)	100%
39	Average (mean) age	100%
243	Median household income (total, in 2020)	100%
1683	Total population in private households	25%
1684	Total visible minority population	25%
1685	Total South Asian population	25%
1686	Total Chinese population	25%
1687	Total Black population	25%
1688	Total Filipino population	25%
1689	Total Arab population	25%
1690	Total Latin American population	25%
1691	Total Southeast Asian population	25%
1692	Total West Asian population	25%
1693	Total Korean population	25%
1694	Total Japanese population	25%
1998	Total population aged 15+	25%
2001	Total population with a postsecondary certificate	25%

Table C3: Structure of missing Census data. Missingness is hierarchical in the Census dataset: entries at a higher stage of missingness are always missing everything in the lower stages.

Missing Stage	Count	% of Total
0. Complete	47593	96.78%
1. Missing Median Household Income	1288	2.62%
2. Missing 25% Sample Variables	57	0.12%
3. Missing Age	211	0.43%
4. Missing Population	26	0.05%

Table C4: Custom census lookup table. This table provides desired socio-demographic information for every dissemination area in the 2021 Census. Only a sample of four entries are shown. Only population, density, and median household income are shown.

Dissemination Area (UID)	**Population**	**Density (per sqkm)**	**Median Household Income (CAD)**
35202387	612	4381.2	$98,500
35061205	548	3920.7	$87,200
59153201	701	6210.5	$76,300
24661048	489	5104.9	$92,100

their removal on our analysis. By the *Canada Elections Act*, donors during a given calendar year (or writ period for donations to candidates) who contributed less than \$200 in total are not required to have their name or address disclosed to Elections Canada (Elections Canada 2026f). Donations whose donors did not disclose this personal information are recorded in our dataset as aggregated sums, grouped by recipient and time-period (writ period for candidates, and quarterly or annual reports for district associations). Otherwise, donations under \$200 appear as individual entries with associated donor names and locations. These two cases are compared against the data separately, and shown in Figure D1, and Figure D2 respectively.

Anonymous contributions under \$200 are the largest source of missing data, and are overall similar in distribution to our study data across the factors considered (Figure D1). A larger proportion of these were addressed to district associations, Liberal entities over Conservative entities, and earlier in the decade (2015 to 2018, and 42nd general election) rather than later (2021 to 2023, and 44th general election). Anonymous contributions were nearly identically distributed to those used in our study with respect to the region of the receiving entity. No single factor among those considered in the anonymous contributions deviated from the study data by more than 10%.

Donor-geolocated contributions under \$200 were distributed differently than the data in our study (Figure D2). A majority (83%) of these contributions were addressed to Liberal district associations specifically, largely outside general election periods (see Panel E in Figure D2). These contributions are similarly distributed to our study data across regions, and in time for those few candidate-bound cases. It is unclear why donors who donate to Liberal district associations are more likely to have their donor locations disclosed.

We also found that only 18% of these donations were sent out-of-district (21.5% of the total amount contributed), compared to 42% and 46% in the studied dataset. While we cannot confirm these numbers for the anonymous contributions, this suggests that contributions under \$200 are overall less likely to leave their districts of origin. This is also supported by the positive association between contributed amount and out-of-district propensity found in our study. While this suggests that our point estimates of out-of-district donation rates are overestimates of the true out-of-district propensity (especially the amount-unweighted proportions), the similarity in distributions between these and contributions above \$200 (especially across regions) suggests that excluding these contributions did not greatly impact our model findings.

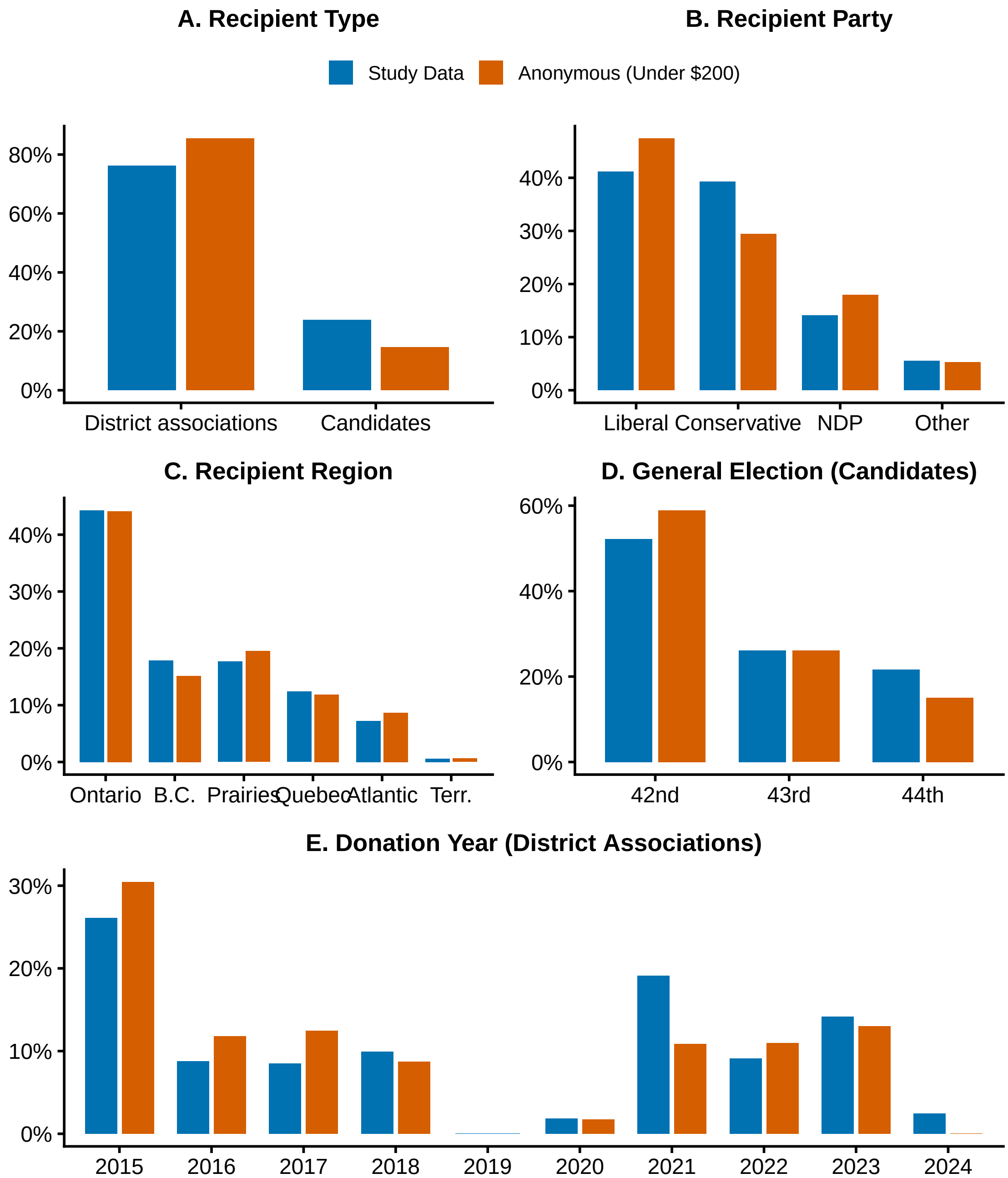


Figure D1: Comparing the distributions of donations to candidates and district associations above $200 whose donors were successfully situated in their districts (blue; those included in our study) vs. anonymous donations under $200 (orange; not included in the study). Datasets are compared across five factors, and distributions are weighted by total amount contributed. Panels D and E only compare datasets for donations bound for candidates and district associations respectively.

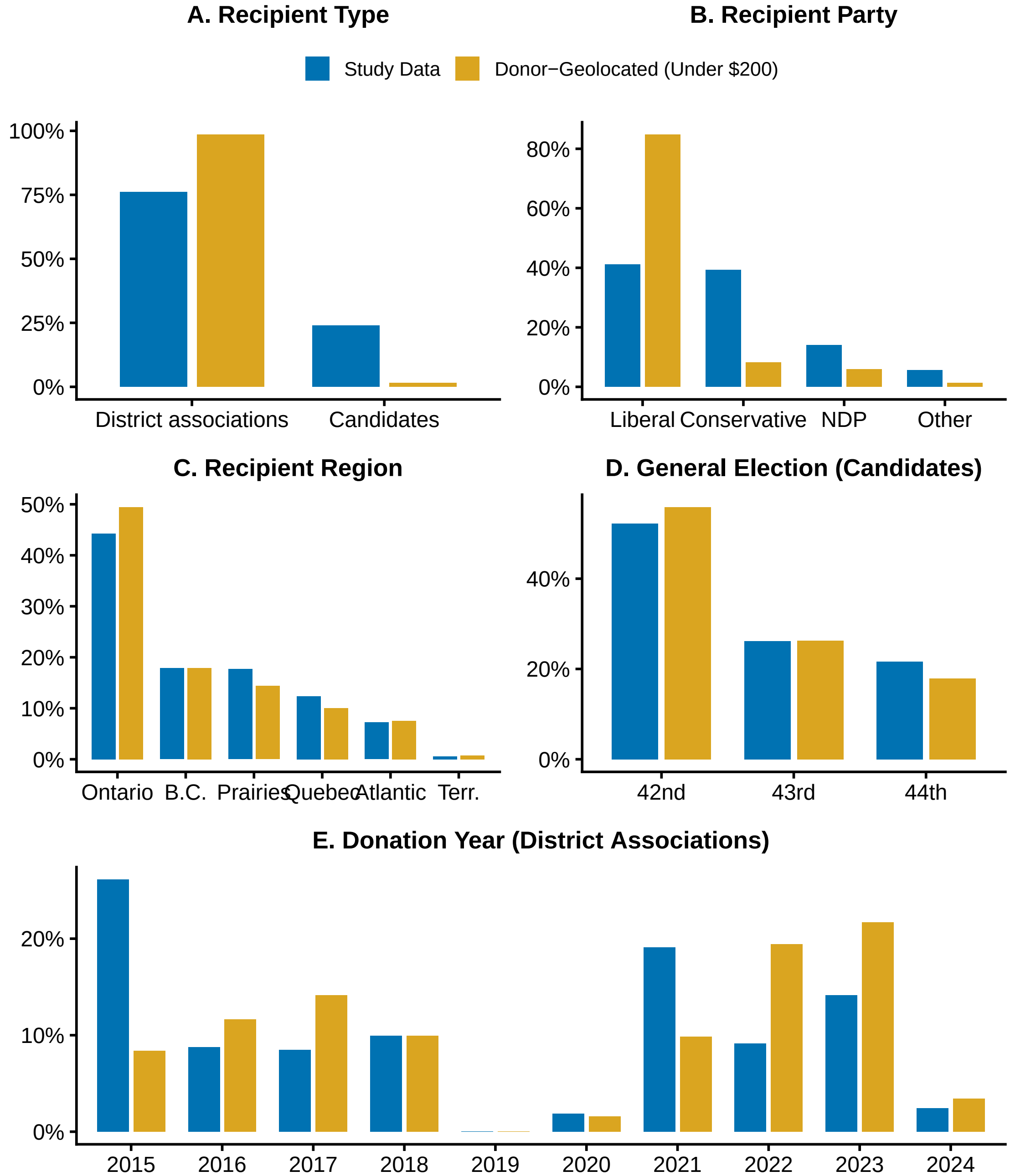


Figure D2: Comparing the distributions of donations to candidates and district associations above $200 whose donors were successfully situated in their districts (blue; those included in our study) vs. location-disclosed donations under $200 (yellow; not included in the study). Datasets are compared across five factors, and distributions are weighted by total amount contributed. Panels D and E only compare datasets for donations bound for candidates and district associations respectively.

Table D1: Breakdown of all contributions addressed to district associations and candidates (during general election periods) from 2015 to 2024, by cause of exclusion from analysis. Study Data is included for reference. We display the share of the total amount received by these entities capture by each case.

Data Case	Share of Total Contribution Amount
Study Data	66.1%
Unsuccessfully Donor-Geolocated (Over $200)	1.6%
Donor-Geolocated (Under $200)	7.9%
Anonymous (Under $200)	24.4%

E Alternate Model Specifications

Here we present the results of six alternate out-of-district propensity models, mainly fit to assess the robustness of our results in Chapter 4 to other resonable model specifications.

Table E1 shows results from an individual-level model with the same covariates, fit using only donations bound for district associations (i.e. excluding those received by candidates). We observe similar coefficients and overall model fit. The only major difference between this and the primary model is that average age is negatively associated with out-of-district propensity, with statistical significance at the $\alpha = 0.05$ level. Table E2 shows results from a district-level model with the same covariates, fit using only donations bound for district associations (i.e. excluding those received by candidates). We observe similar coefficients and overall model fit.

Table E3 shows results from an individual-level model with the same covariates, fit using only donations received by candidates (i.e. excluding those received by district associations). We observe similar coefficient and overall model fit. Table E4 shows results from a slighly more elaborate version of this model, that includes the election period in which the donation was received, as well as how competitive the donor's district was during that period. Both of these more specific covariates were significantly assocated with out-of-district propensity, but did not affect the rough magnitudes and significances anywhere else in the model.

Table E5 and Table E6 shows results from individual level and district level models respectively, where province and visible minority proportion were disaggregated into more specific categories. In both models, the coefficients associated with provinces each behaved similarly to their parent region. The proportion of habitants belonging to all ethnic groups available in the Census were positively assocaited with out-of-district donation at the individual level. However, only the proportion of South Asian, Black, and Chinese inhabitants of a district was associated with it's share of donations that were bound for other districts. Further, disaggregation greatly increased the number of ocefficients used to fit the model, for marginal gains in the model's explanatory power. For example, the disaggregated district level model has over double the number of coefficients compared to the one included in the main text, for only a single percentage increase in the model's explainability. For this reason, grouped variables were kept in the main models at both levels.

F Testing Whether Out-of-District Rates can be Explained by Distance Alone

F.1 Simulation Setup

Figure 3 shows how variable out-of-district donation rates are at the district level. However, this may simply be a byproduct of the spatial arrangement of electoral districts. Districts vary enormously in size, from the 13 square kilometres that make up Toronto–St. Paul's, to the entire territory of Nunavut. Assuming that social influence decays as a function of distance, we should expect a donor residing in a district in downtown Toronto to (1) be more aware of or (2) perceive to have a greater stake in district associations and candidates outside their own district, simply by virtue of their spatial proximity. By contrast, a donor in Nunavut may be expected to place less importance on political entities in the neighbouring Northwest Territories, whose population centre is nearly one thousand kilometres away. In this section we design a simulation to test the null hypothesis that the observed variance in district-level out-of-district donation rates is well-explained by differences in spatial clustering.

- Assign a random distance to each donation.
- If this distance is < 1km, immediately declare the donation as having been received in-district. Otherwise, continue.
- Assign a random direction for that donation.
- Determine if it landed within Canada. If it didn't choose a different direction. Otherwise, continue.
- Determine which district it landed in.
- Declare in-district vs. out-of-district.

Distances were assigned by sampling from the true distance distribution of the observed donations dataset, shown in Figure F1 (we note that this figure only shows out-of-district donations; the majority of donations were in-district, and therefore travelled zero kilometres). The distance a donation travelled was defined to be zero if the donation was in-district, and otherwise set to be the spatial distance between the centroid of the donor's dissemination area,

Table E1: Results from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. Only donations sent to district assocaitions are considered. The reference party is the Conservative Party, and the reference region is Ontario. Estimates are bolded when significant at the alpha = 0.05 level. Small losses (~ 1.1% of entries) were incurred due to missingness in the census data.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−2.789**	**0.069**	**0.062**	**0.004**
Conservative	-		-	
Liberal	**0.367**	**0.016**	**1.443**	**0.023**
N.D.P	0.020	0.022	1.021	0.022
Other	**0.153**	**0.038**	**1.166**	**0.044**
Amount	**0.000**	**0.000**	**1.000**	**0.000**
Ontario	-		-	
Quebec	**0.294**	**0.023**	**1.342**	**0.030**
British Columbia	**−0.427**	**0.019**	**0.653**	**0.013**
Prairies	**−0.419**	**0.021**	**0.658**	**0.014**
Atlantic Canada	**−0.254**	**0.032**	**0.776**	**0.025**
Territories	**−1.842**	**0.152**	**0.159**	**0.024**
Age	**−0.004**	**0.001**	**0.996**	**0.001**
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.890**	**0.066**	**6.617**	**0.438**
Visible Minority	**2.054**	**0.032**	**7.801**	**0.252**
margin_of_victory_avg	**0.008**	**0.001**	**1.008**	**0.001**
Population Density (sqrt)	0.003	0.000	1.003	0.000
Num.Obs.	107 442		107 442	
AIC	130 829.1		130 829.1	
BIC	130 982.5		130 982.5	
Log.Lik.	−65 398.556		−65 398.556	
F	914.461		914.461	
RMSE	0.46		0.46	

Table E2: Results from a beta regression model where the outcome is the out-of-district share of the sum of EDA-bound contributions over $200 originating from an individual district during our study period. The reference region is Ontario. Estimates are bolded when significant at the alpha = 0.05 level. Small losses (~ 1.1% of individual transactions) were incurred due to missingness in the census data. Each of the 338 districts is represented in this model.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−3.939**	**0.862**	**0.019**	**0.017**
Amount	0.000	0.000	1.000	0.000
Ontario	-		-	
Quebec	**0.538**	**0.120**	**1.713**	**0.205**
British Columbia	−0.182	0.104	0.834	0.087
Prairies	−0.205	0.117	0.815	0.095
Atlantic Canada	**0.301**	**0.151**	**1.351**	**0.203**
Territories	**−1.222**	**0.497**	**0.295**	**0.147**
Age	0.012	0.017	1.012	0.017
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.658**	**0.646**	**5.247**	**3.391**
Visible Minority	**1.847**	**0.238**	**6.340**	**1.511**
Margin of Victory (mean)	0.004	0.003	1.004	0.003
Population Density (sqrt)	**0.012**	**0.002**	**1.012**	**0.002**
Num.Obs.	338		338	
R2	0.607		0.607	
AIC	−401.2		−401.2	
BIC	−343.8		−343.8	
RMSE	0.14		0.14	

Table E3: Results from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. Only donations sent to electoral candidates during election periods are considered. The reference party is the Conservative Party, the reference region is Ontario. Estimates are bolded when significant at the alpha = 0.05 level. Small losses (~ 1.1% of entries) were incurred due to missingness in the census data.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−3.450**	**0.143**	**0.032**	**0.005**
Conservative	-		-	
Liberal	**0.079**	**0.037**	**1.082**	**0.040**
N.D.P	0.007	0.036	1.007	0.036
Other	**0.618**	**0.045**	**1.856**	**0.083**
Amount	**0.000**	**0.000**	**1.000**	**0.000**
Ontario	-		-	
Quebec	**0.398**	**0.048**	**1.489**	**0.071**
British Columbia	**−0.438**	**0.043**	**0.645**	**0.028**
Prairies	**−0.292**	**0.044**	**0.747**	**0.033**
Atlantic Canada	**−0.237**	**0.051**	**0.789**	**0.040**
Territories	**−3.012**	**0.507**	**0.049**	**0.025**
Age	0.005	0.002	1.005	0.002
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**2.011**	**0.136**	**7.471**	**1.019**
Visible Minority	**2.636**	**0.063**	**13.960**	**0.878**
margin_of_victory_avg	**0.005**	**0.001**	**1.005**	**0.001**
Population Density (sqrt)	0.004	0.001	1.004	0.001
Num.Obs.	28 995		28 995	
AIC	33 320.5		33 320.5	
BIC	33 452.9		33 452.9	
Log.Lik.	−16 644.236		−16 644.236	
F	273.999		273.999	
RMSE	0.44		0.44	

Table E4: Results from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. Only donations sent to electoral candidates during election periods are considered. The reference party is the Conservative Party, the reference region is Ontario, and the reference general election period is the 42nd. Estimates are bolded when significant at the alpha = 0.05 level. Small losses (~ 1.1% of entries) were incurred due to missingness in the census data.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−3.339**	**0.143**	**0.035**	**0.005**
Conservative	-		-	
Liberal	0.059	0.037	1.061	0.039
N.D.P	−0.014	0.036	0.986	0.036
Other	**0.638**	**0.045**	**1.893**	**0.085**
42nd general election	-		-	
43rd general election	**−0.158**	**0.033**	**0.854**	**0.028**
44th general election	**−0.119**	**0.036**	**0.888**	**0.032**
Amount	**0.000**	**0.000**	**1.000**	**0.000**
Ontario	-		-	
Quebec	**0.402**	**0.047**	**1.495**	**0.071**
British Columbia	**−0.449**	**0.043**	**0.638**	**0.028**
Prairies	**−0.260**	**0.042**	**0.771**	**0.032**
Atlantic Canada	**−0.224**	**0.052**	**0.799**	**0.041**
Territories	**−3.028**	**0.507**	**0.048**	**0.025**
Age	0.005	0.002	1.005	0.002
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.989**	**0.136**	**7.307**	**0.995**
Visible Minority	**2.636**	**0.063**	**13.950**	**0.879**
Margin of Victory (per-cycle)	**0.003**	**0.001**	**1.003**	**0.001**
Population Density (sqrt)	**0.004**	**0.001**	**1.004**	**0.001**
Num.Obs.	28 995		28 995	
AIC	33 304.4		33 304.4	
BIC	33 453.3		33 453.3	
Log.Lik.	−16 634.198		−16 634.198	
F	242.454		242.454	
RMSE	0.44		0.44	

Table E5: Results from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. The reference entity is district associations, the reference party is the Conservative Party, the reference province is Ontario. Estimates are bolded when significant at the alpha = 0.05 level. Small losses (~ 1.1% of entries) were incurred due to missingness in the census data.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−3.323**	**0.067**	**0.036**	**0.002**
Conservative	-		-	
Liberal	**0.328**	**0.014**	**1.388**	**0.020**
N.D.P	0.012	0.019	1.012	0.019
Other	**0.344**	**0.028**	**1.410**	**0.040**
political_entityRegistered associations	**-**		**-**	
political_entityCandidates	−0.141	0.015	0.868	0.013
Amount	0.000	0.000	1.000	0.000
donor_provinceAlberta	-		-	
donor_provinceBritish Columbia	0.043	0.027	1.044	0.029
donor_provinceManitoba	0.167	0.037	1.181	0.043
donor_provinceNew Brunswick	0.063	0.051	1.065	0.054
donor_provinceNewfoundland and Labrador	0.723	0.054	2.061	0.111
donor_provinceNorthwest Territories	**−1.193**	**0.260**	**0.303**	**0.079**
donor_provinceNova Scotia	**0.045**	**0.045**	**1.046**	**0.047**
donor_provinceNunavut	−1.834	0.423	0.160	0.068
donor_provinceOntario	**0.472**	**0.023**	**1.603**	**0.037**
donor_provincePrince Edward Island	0.099	0.072	1.104	0.079
donor_provinceQuebec	0.723	0.028	2.060	0.057
donor_provinceSaskatchewan	0.195	0.040	1.216	0.049
donor_provinceYukon	−1.638	0.194	0.194	0.038
Age	−0.001	0.001	0.999	0.001
Income	0.000	0.000	1.000	0.000
Education	1.759	0.064	5.807	0.369
vismin_south_asian	1.988	0.045	7.304	0.325
vismin_chinese	2.411	0.056	11.144	0.627
vismin_black	2.533	0.147	12.596	1.851
vismin_filipino	1.627	0.154	5.090	0.781
vismin_arab	2.192	0.194	8.952	1.737
vismin_latin_american	6.726	0.330	833.532	275.452
vismin_southeast_asian	3.916	0.341	50.222	17.134
vismin_west_asian	1.421	0.226	4.140	0.936
vismin_korean	2.514	0.393	12.354	4.860
vismin_japanese	3.253	0.824	25.859	21.304
margin_of_victory_avg	0.008	0.001	1.008	0.001

Table E6: Results from a beta regression model where the outcome is the out-of-district share of the sum of contributions over $200 originating from an individual district during our study period. The reference province is Ontario. Estimates are bolded when significant at the alpha = 0.05 level. Small losses (~ 1.1% of individual transactions) were incurred due to missingness in the census data. Each of the 338 districts is represented in this model.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−5.164**	**0.847**	**0.006**	**0.005**
Amount	0.000	0.000	1.000	0.000
donor_provinceAlberta	-		-	
donor_provinceBritish Columbia	0.177	0.177	1.194	0.211
donor_provinceManitoba	0.379	0.184	1.461	0.269
donor_provinceNew Brunswick	0.546	0.233	1.725	0.401
donor_provinceNewfoundland and Labrador	0.756	0.257	2.130	0.547
donor_provinceNorthwest Territories	−0.896	0.775	0.408	0.316
donor_provinceNova Scotia	0.395	0.220	1.484	0.327
donor_provinceNunavut	**−0.469**	**0.969**	**0.626**	**0.606**
donor_provinceOntario	**0.238**	**0.134**	**1.269**	**0.170**
donor_provincePrince Edward Island	0.200	0.327	1.221	0.400
donor_provinceQuebec	0.722	0.157	2.058	0.323
donor_provinceSaskatchewan	**0.409**	**0.187**	**1.506**	**0.281**
donor_provinceYukon	−1.395	0.776	0.248	0.192
Age	0.023	0.017	1.023	0.017
Income	0.000	0.000	1.000	0.000
Education	1.977	0.722	7.218	5.212
vismin_south_asian	1.562	0.411	4.771	1.961
vismin_chinese	2.250	0.444	9.484	4.215
vismin_black	4.204	1.206	66.976	80.740
vismin_filipino	0.653	1.191	1.921	2.287
vismin_arab	2.687	1.567	14.687	23.011
vismin_latin_american	3.856	4.289	47.256	202.694
vismin_southeast_asian	6.267	4.493	527.095	2368.115
vismin_west_asian	0.402	2.303	1.494	3.442
vismin_korean	4.480	4.205	88.201	370.871
vismin_japanese	−6.419	17.602	0.002	0.029
Margin of Victory (mean)	0.004	0.003	1.004	0.003
Population Density (sqrt)	0.010	0.003	1.010	0.003
Num.Obs.	338		338	
R2	0.683		0.683	
AIC	−460.0		−460.0	
BIC	−341.5		−341.5	
RMSE	0.12		0.12	

and the centroid of the recipient's district. Shapefiles were used to determine which district a randomly travelling donation landed in.

For each district, 30 replicate simulations were run to estimate the sample distribution of the simulation. Using these distributions, the observed out-of-district proportions were compared with the simulated out-of-district proportions, for each district. Only results for the count of out-of-district donors (not the amount of donations sent out-of-district) was considered, however results were very similar for amounts, just with systematically wider sampling distributions.

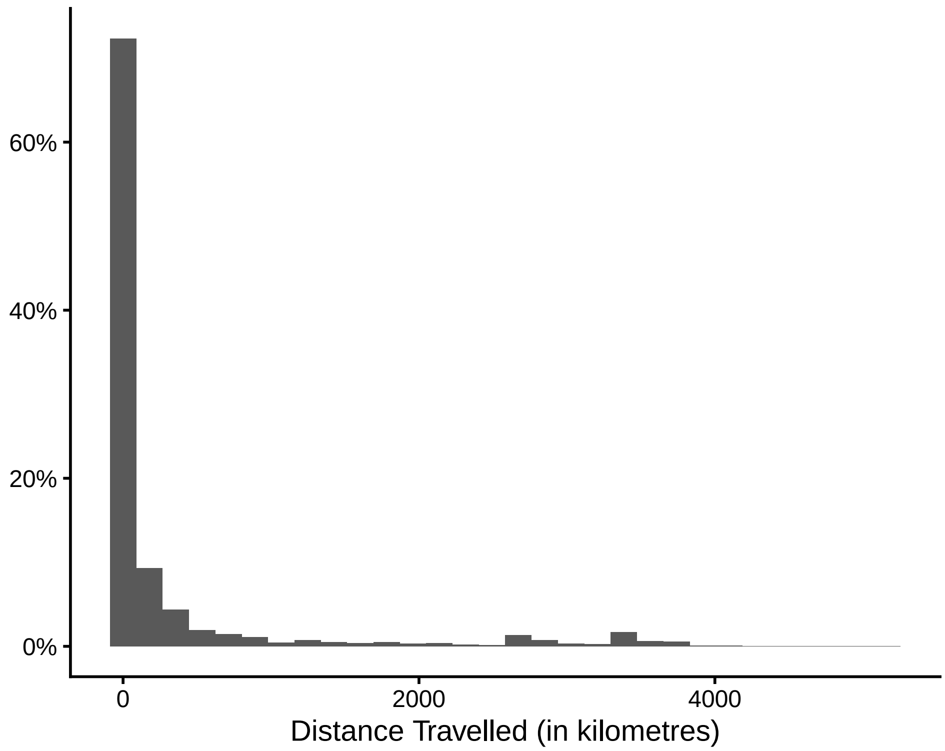


Figure F1: Distribution of donation distances for out-of-district donations over $200 during our study period. Both donations to candidates and district associations are shown together, however distributions were very similar for both entities separately. Distance is measured as the distance from the donor dissemination area's centroid to the recipient district's centroid.

F.2 Simulation Results

When considering donations addressed to candidates (aggregated over all three election periods), only 6.5% of districts fell within the expected proportion of out-of-district donations. For donations addressed to EDAs over the entire study period, only 5.3% of districts fell within the expected proportion of out-of-district donations. Proportions of districts that fell within

Table G1: Summaries of financial contributions to federal political entities over $200 from 2015 to 2024, separated by entity. Only donations stated to have been received during a general election period are shown. Contributions received by nomination contestants are not shown.

Political Entity	Type	Donation Count	Donation Amount (Thousands)	Share of Total Amount
Candidate	Localized	29,380	$18,872	25.7%
District Association	Localized	36,489	$20,373	27.7%
Leadership Contestant	National	134	$103	0.1%
Party	National	79,052	$34,071	46.4%

their expected ranges were slightly higher (8.2% and 8.5% respectively) when looking at the share of the total contributed amount, however this was likely due to the systematically wider sampling distributions caused by the added variance incurred when using amounts.

Figure F2 shows that, for both entities, most districts have out-of-district proportions that deviate drastically from simulated expectations under the null hypothesis. Further, most deviations, including the largest ones in magnitude, were districts where the observed out-of-district donation rate was larger than expected, as opposed to smaller. This indicates that most districts have higher out-of-district donation rates than what can be explained by the spatial arrangement of districts alone. This can be seen more clearly in Figure F3; in both cases, much of the support for the observed distribution lies strictly above the support for the expected distribution. This skew in deviations is especially noticeable for donations addressed to district associations. Results using amount-weighted proportions were similar in both cases.

G Additional Data Summaries

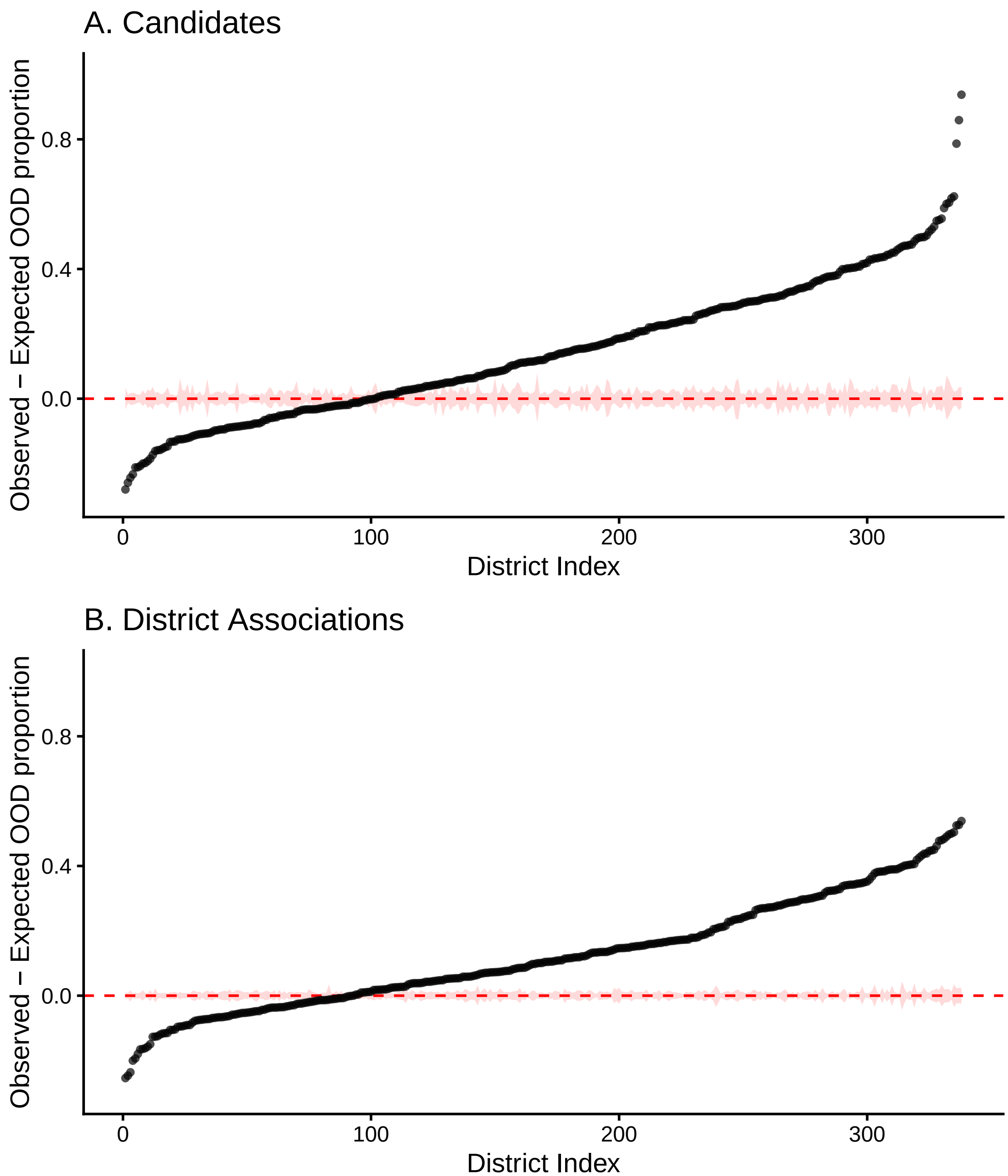


Figure F2: Deviation of each district's observed proportion of out-of-district (OOD) donations from its simulated proportion under the null hypothesis. Results for both donations to candidates (A) and district associations (B) are shown. Districts are sorted by the sign and magnitude of their deviation. The dashed red line indicates no deviation, and the shaded red band gives the simulated 95% confidence interval. Points outside the band deviate more than spatial clustering alone can explain.

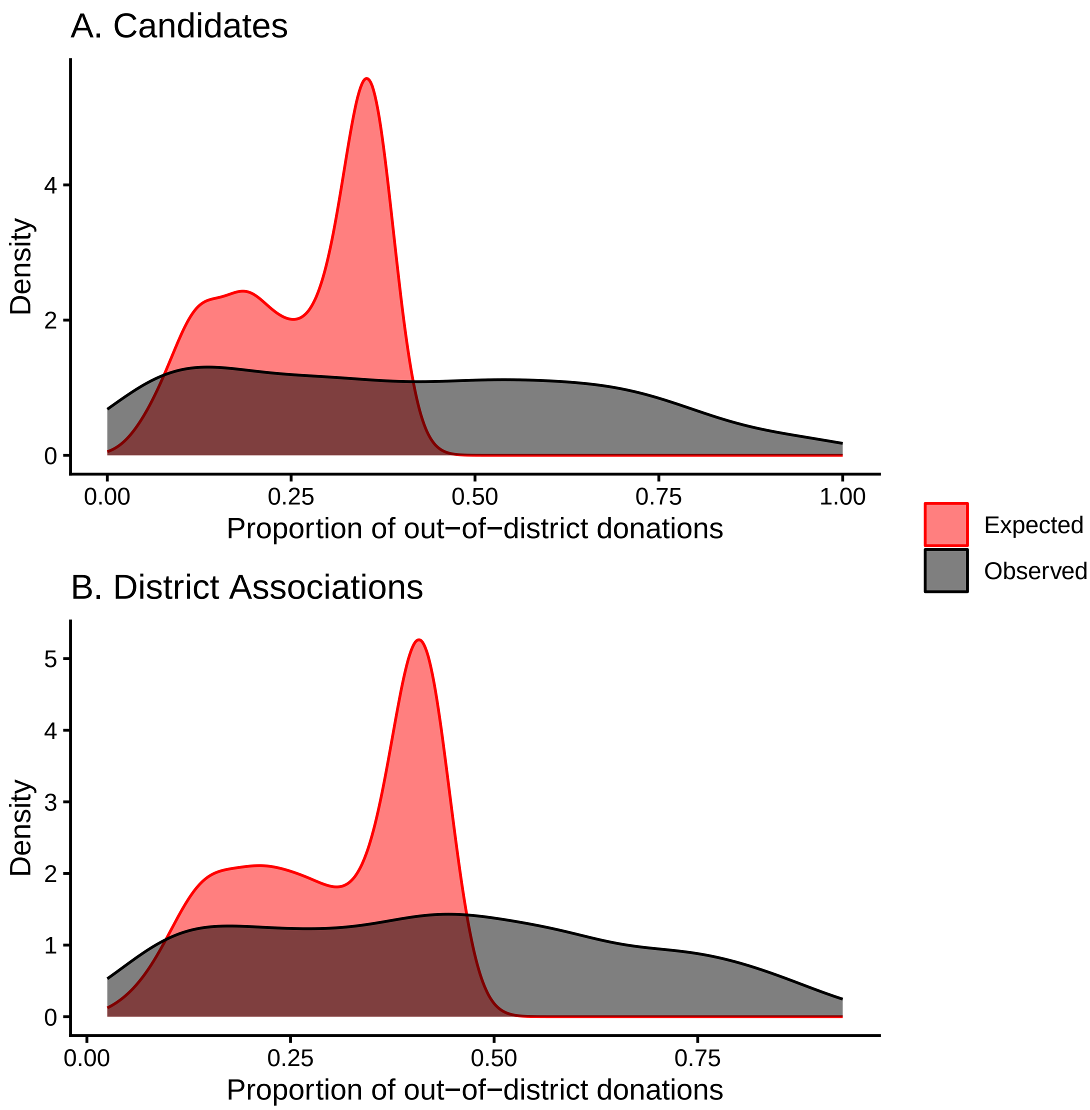


Figure F3: Distribution across districts of the observed (black) and simulated null-expected (red) proportion of out-of-district donations. Both donations to candidates (A) and district associations (B) are shown.

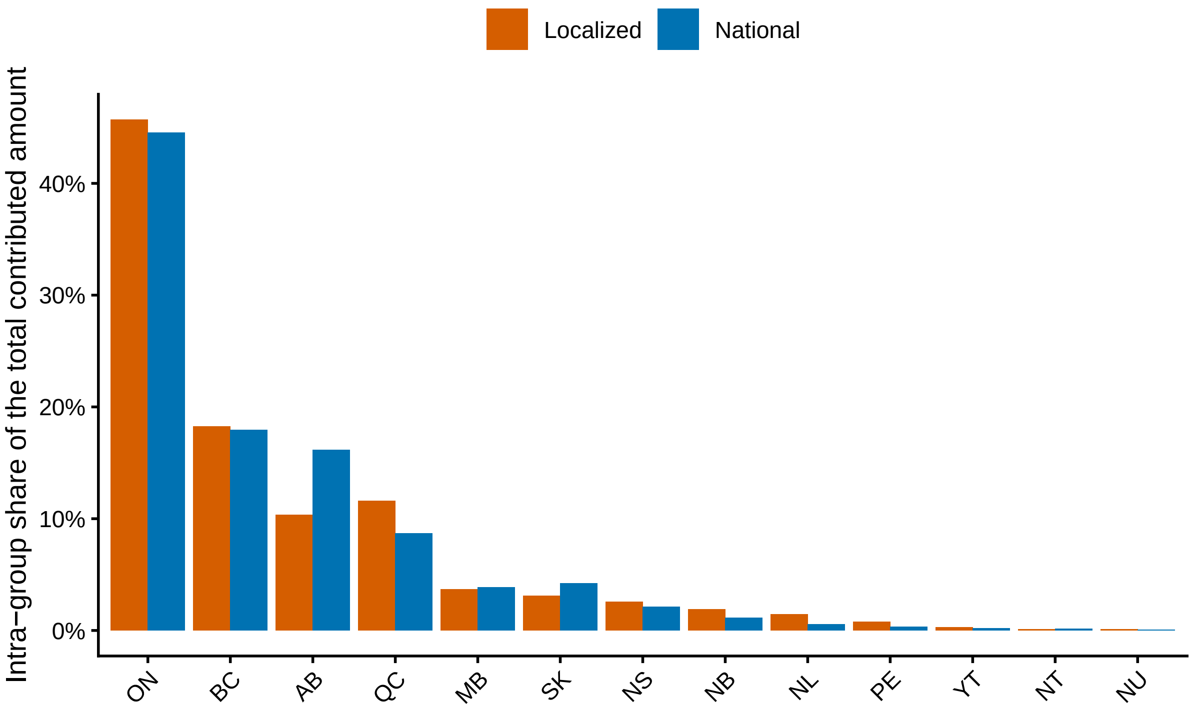


Figure G1: Distribution of contributions across donor provinces, shown separately for localized entities (candidates and district associations) and national ones (parties and leadership contestants). Within each group, bars give each province's share of that group's total contributed amount, and sum to 100% across provinces. Contributions over $200 received from 2015 to 2024.

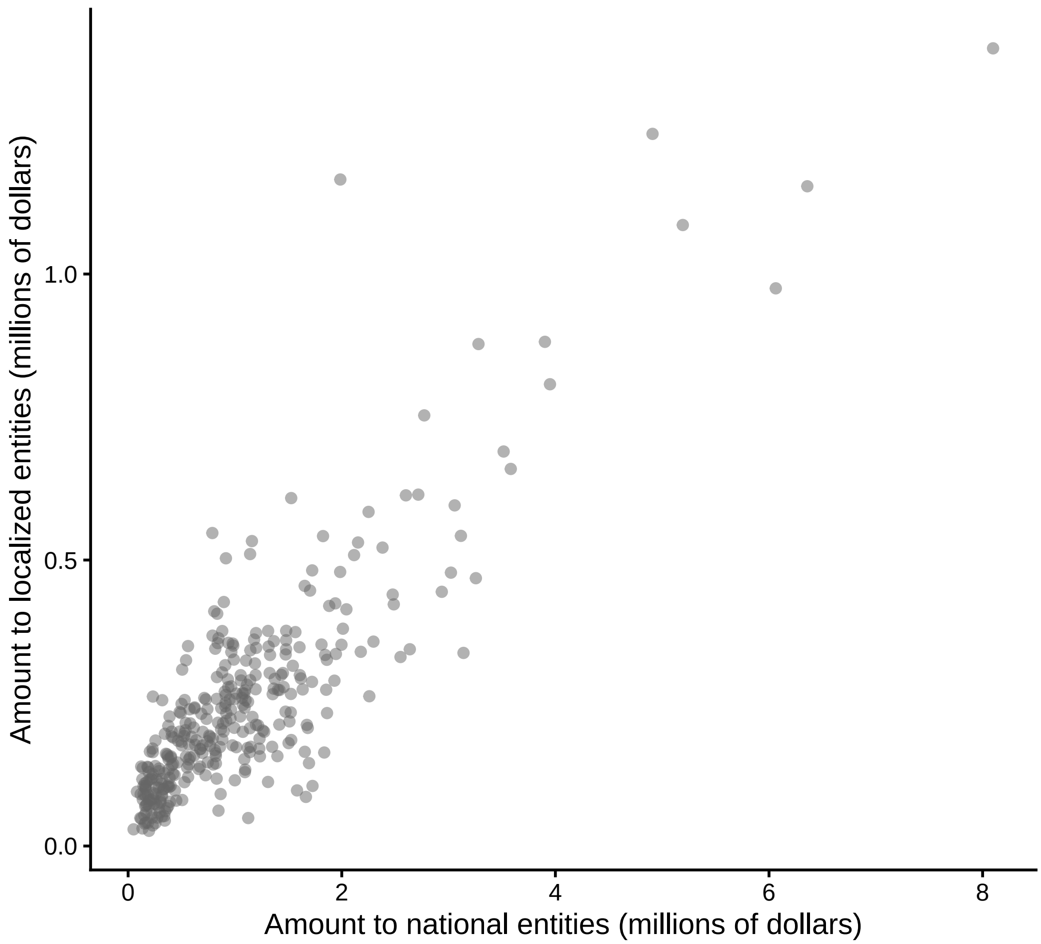


Figure G2: Total amount contributed to localized entities (candidates and district associations) against the total contributed to national entities (parties and leadership contestants), with one point per donor district. Contributions over $200 received from 2015 to 2024.

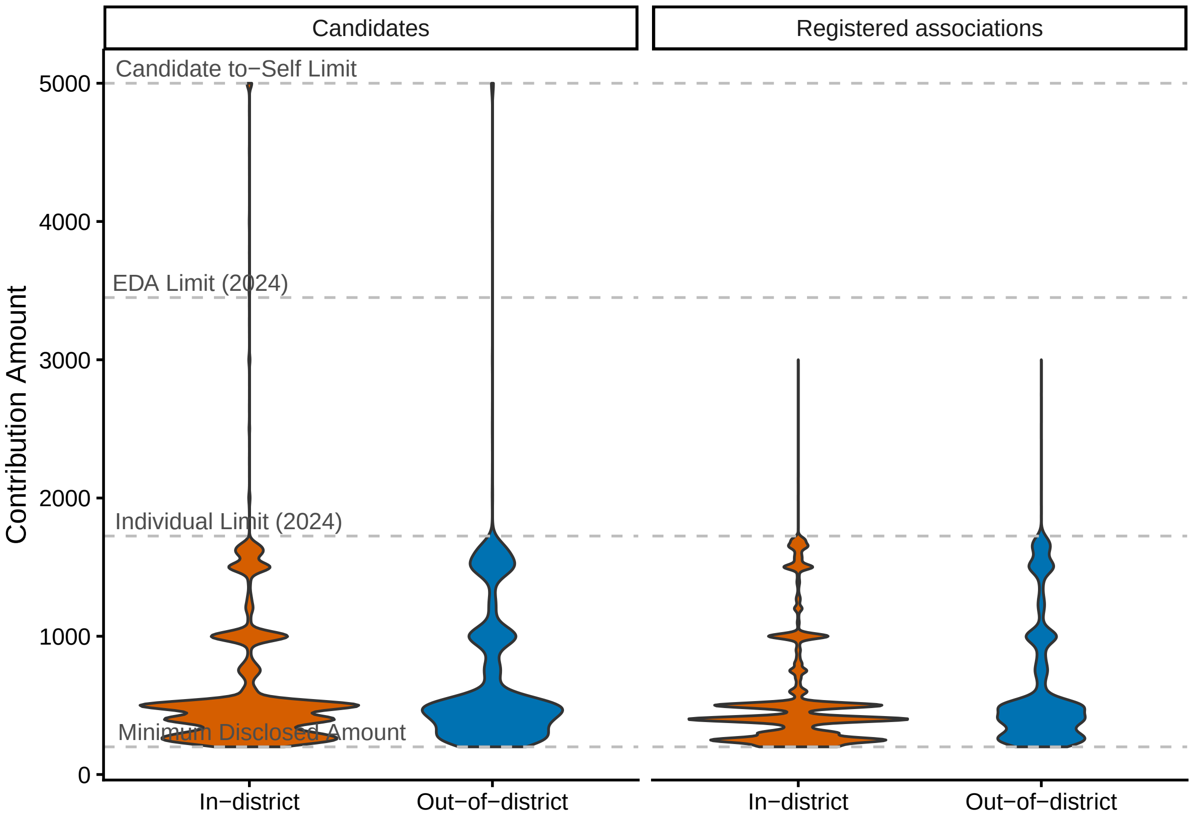


Figure G3: Value distributions of in- (blue) and out-of-district (red) contributions over \$200 received by candidates (left) and district associations (right) from 2015 to 2024. Dashed grey lines mark notable individual contribution limits.

Table G2: Proportion of contributions over $200 that were received by localized entities (candidates and district associations) outside the donor's district, from 2015 to 2024, broken down by recipient entity and party. The three largest parties are shown individually; all others are grouped into 'Other'. Raw and value-weighted proportions of out-of-district donations (OOD) are reported, alongside overall contribution counts for reference. Contributions received by candidates outside general election cycles are not included in calculations.

Entity	Party	Donation Count	OOD Proportion	OOD Proportion (Amount Weighted)
Candidate	Conservative Party of Canada	14,421	36.1%	38.5%
Candidate	Liberal Party of Canada	5,961	39.7%	43.9%
Candidate	New Democratic Party	6,637	35.9%	39.6%
Candidate	Other	3,540	43.3%	41.4%
District Association	Conservative Party of Canada	38,756	36.0%	40.1%
District Association	Liberal Party of Canada	49,859	53.5%	57.8%
District Association	New Democratic Party	16,164	35.2%	35.8%
District Association	Other	3,840	34.6%	36.0%

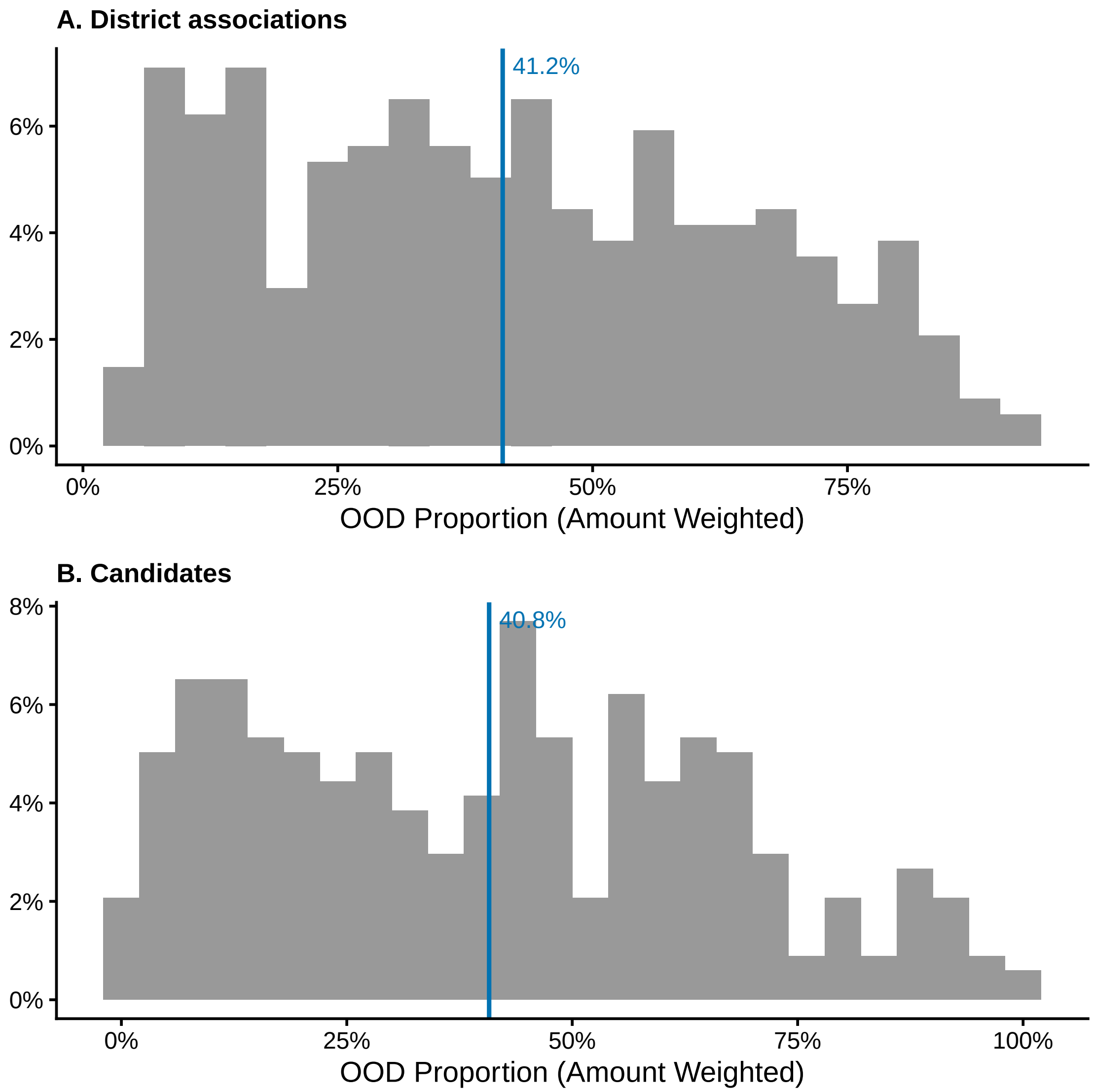


Figure G4: Distribution of out-of-district (OOD) contribution shares for each district in our study (for contributions over $200, from 2015 to 2024). (A) and (B) show this distribution for donations to district associations and candidates separately. The district-level mean is indicated with an orange vertical line in each case. Distributions were similar when raw instead of amount-weighted proportions were used.

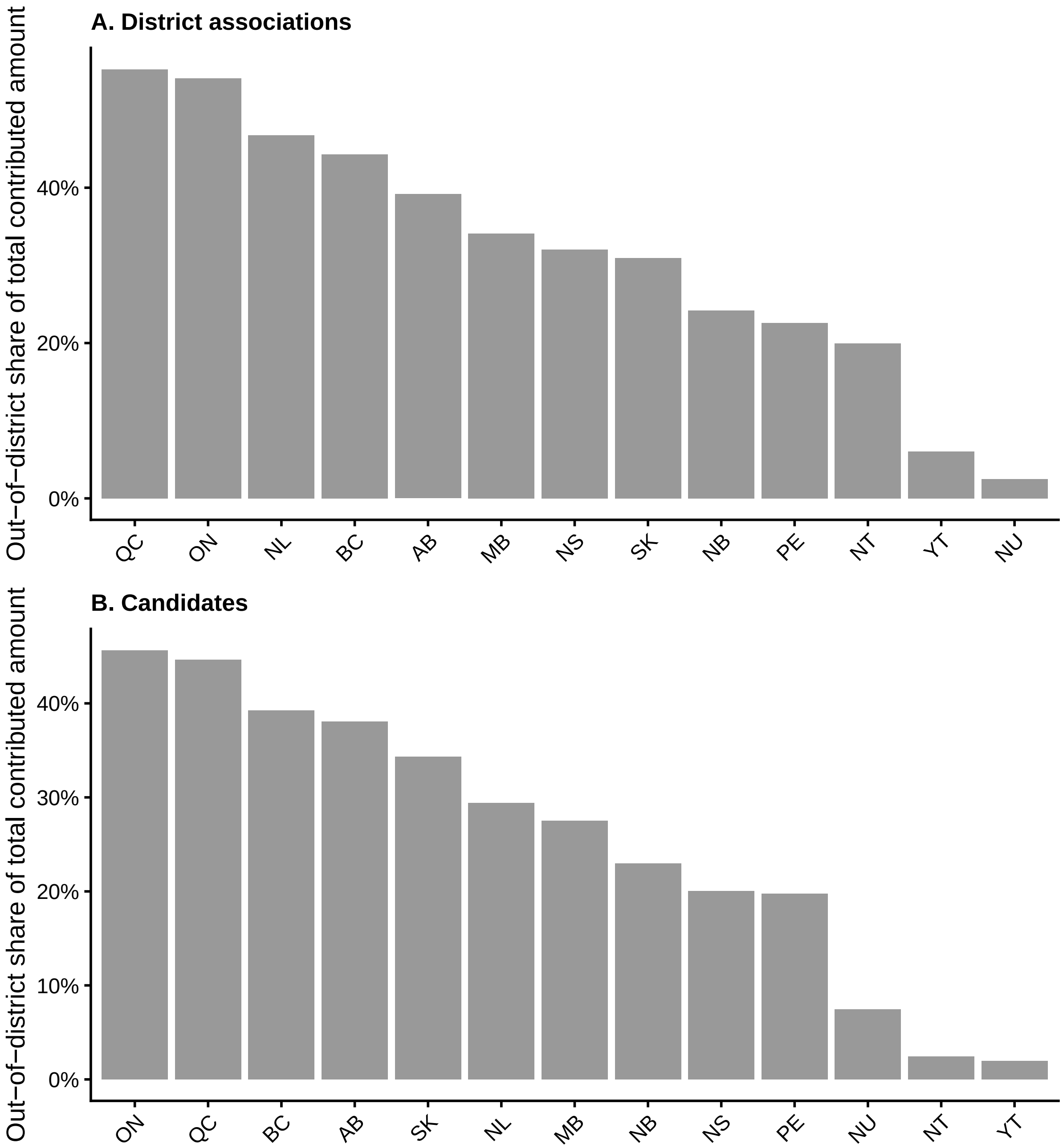


Figure G5: Out-of-district share of the total amount contributed over $200 to (A) district associations and (B) candidates, for each donor province, from 2015 to 2024. Within each panel, provinces are ordered from highest to lowest out-of-district share.

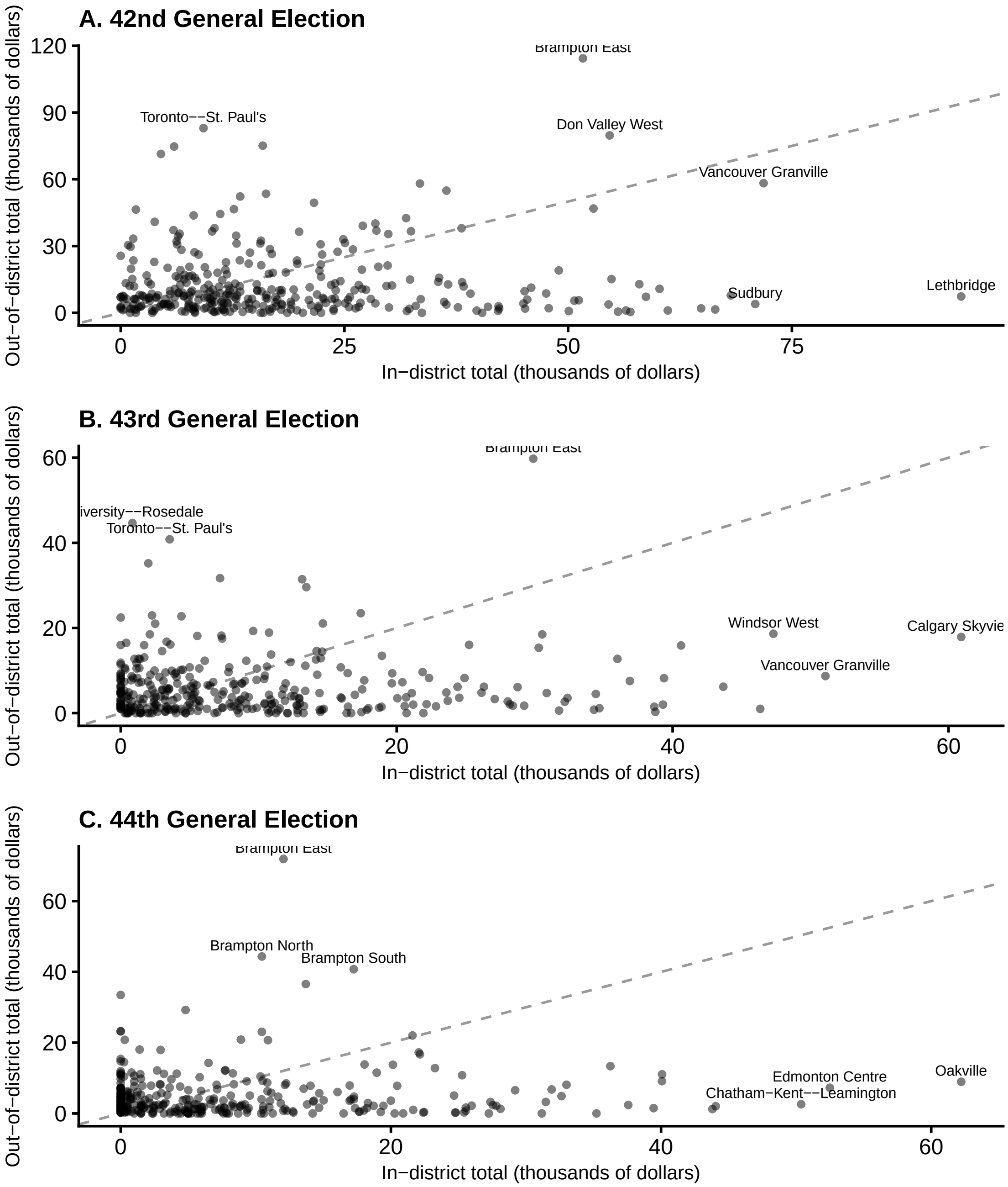


Figure G6: Relationship between the sum of contributions over $200 donated in- v.s. out-of-district, for each district during the 2013 representational order (effective between 2015 and 2024). (A), (B), and (C) display this relationship for donations to candidates during the 42nd, 43rd, and 44th writ periods respectively. Points represent individual districts. The grey dashed line is 1:1. The top 3 districts by amount contributed in- and out-of-district respectively are labelled with their names.

References

Ansolabehere, Stephen, John M. de Figueiredo, and James M. Snyder Jr. 2003. "Why Is There so Little Money in U.S. Politics?" *Journal of Economic Perspectives* 17 (1): 105–30. https://doi.org/10.1257/089533003321164976.

Baker, Anne E. 2016. "Getting Short-Changed? The Impact of Outside Money on District Representation." *Social Science Quarterly* 97 (5): 1096–107. https://www.jstor.org/stable/26612374.

Bednar, Jenna, and Elizabeth R. Gerber. 2011. *Political Geography, Campaign Contributions, and Representation*. https://public.websites.umich.edu/~jbednar/WIP/irod_050211_wfigs.pdf.

Besco, Randy. 2019. *Identities and Interests: Race, Ethnicity, and Affinity Voting*. UBC Press. https://www.ubcpress.ca/identities-and-interests.

Besco, Randy, and Erin Tolley. 2022. "Ethnic Group Differences in Donations to Electoral Candidates." *Journal of Ethnic and Migration Studies* 48 (5): 1072–94. https://doi.org/10.1080/1369183X.2020.1804339.

Bramlett, Brittany H., James G. Gimpel, and Frances E. Lee. 2011. "The Political Ecology of Opinion in Big-Donor Neighborhoods." *Political Behavior* 33 (4): 565–600. https://doi.org/10.1007/s11109-010-9144-7.

Canada Post. 2026. *Addressing Guidelines: Overview*. https://www.canadapost-postescanada.ca/cpc/en/support/articles/addressing-guidelines/overview.page#1414221.

Canes-Wrone, Brandice, and Kenneth M. Miller. 2022. "Out-of-District Contributors and Representation in the US House." *Legislative Studies Quarterly*. https://csap.yale.edu/sites/default/files/files/apppw-bcwrevised-2-5-20.pdf.

Carmichael, Brianna, and Paul Howe. 2014. "Political Donations and Democratic Equality in Canada." *Canadian Parliamentary Review*. http://www.revparl.ca/37/1/37n1e_14_carmichael.pdf.

Corrado, Anthony, Thomas E. Mann, Daniel R. Ortiz, and Trevor Potter. 2010. *The New*

Campaign Finance Sourcebook. Brookings Institution Press.

Cribari-Neto, Francisco, and Achim Zeileis. 2010. "Beta Regression in r." *Journal of Statistical Software* 34 (2). https://doi.org/10.18637/jss.v034.i02.

Eagles, Munroe. 2004. "The Effectiveness of Local Campaign Spending in the 1993 and 1997 Federal Elections in Canada." *Canadian Journal of Political Science / Revue Canadienne de Science Politique* 37 (1): 117–36. https://www.jstor.org/stable/25165603.

Elections Canada. 2023. *Implementation of New Federal Electoral Boundaries*. Government of Canada. https://www.elections.ca/content.aspx?section=med&dir=pre&document=sep2723&lang=e.

Elections Canada. 2025a. *Open Data*. https://www.elections.ca/content.aspx?section=fin&dir=oda&document=index&lang=e.

Elections Canada. 2025b. *Redistribution of Federal Electoral Districts 2022*. https://www.elections.ca/content.aspx?section=res&dir=cir/red&document=index&lang=e.

Elections Canada. 2026a. *Annual Limits on Contributions – 2026*. https://www.elections.ca/content.aspx?section=pol&dir=lim&document=lim2026&lang=e.

Elections Canada. 2026b. *Expenses Limits*. https://www.elections.ca/content.aspx?section=pol&dir=limits&document=index&lang=e.

Elections Canada. 2026c. *Financial Returns*. https://www.elections.ca/content.aspx?section=fin&dir=rev&document=index&lang=e.

Elections Canada. 2026d. *Political Entities*. https://www.elections.ca/content.aspx?section=pol&document=index&lang=e.

Elections Canada. 2026e. *Political Financing*. https://www.elections.ca/content.aspx?section=fin&document=index&lang=e.

Elections Canada. 2026f. *Understanding Contributions*. https://www.elections.ca/content.aspx?section=fin&dir=cont&document=index&lang=e.

Federal Election Commission. 2026. "Contribution Limits." https://www.fec.gov/help-candidates-and-committees/candidate-taking-receipts/contribution-limits/.

Garnett, Holly Ann. 2024. "Where Do Donors Come from? Using Census Data to Predict Donations to Canadian Federal Election Candidates." *Political Geography* 110 (April): 103077. https://doi.org/10.1016/j.polgeo.2024.103077.

Garnett, Holly Ann, Scott Pruysers, Lisa Young, and William P. Cross. 2022. "Lifeblood of the Party: Motivations for Political Donations in Canada." *American Review of Canadian Studies* 52 (4). https://doi.org/10.1080/02722011.2022.2147756.

Gimpel, James G., Frances E. Lee, and Joshua Kaminski. 2006. "The Political Geography of Campaign Contributions in American Politics." *The Journal of Politics* 68 (3): 626–39. https://doi.org/10.1111/j.1468-2508.2006.00450.x.

Gimpel, James G., Frances E. Lee, and Shanna Pearson-Merkowitz. 2008. "The Check Is in the Mail: Interdistrict Funding Flows in Congressional Elections." *American Journal of Political Science* 52 (2): 373–94. https://www.jstor.org/stable/25193819.

Hou, Feng. 2004. *Recent Immigration and the Formation of Visible Minority Neighbourhoods in Canada's Large Cities*. Analytical Studies Branch Research Paper Series 11F0019MIE No. 221. Statistics Canada. https://www150.statcan.gc.ca/n1/pub/11f0019m/11f0019m2004221-eng.pdf.

Jacobs, Nicholas, and Wasike Gil Imboywa. 2024. "The Nationalization of Individual Campaign Contributions in U.S. Senate Elections, 1984-2020." *American Politics Research* 52 (3): 239–48. https://doi.org/10.1177/1532673X231220639.

Jansen, Harold J., Melanee Thomas, and Lisa Young. 2012. "Who Donates to Canada's Political Parties?" *Annual Meeting of the Canadian Political Science Association* (Edmonton, Alberta), June. https://www.cpsa-acsp.ca/papers-2012/Jansen-Thomas-Young.pdf.

Keena, Alex. 2024. "Out-Of-State Donors and Legislative Surrogacy in the U.S. Senate." *Political Research Quarterly* 77 (3): 880–90. https://doi.org/10.1177/10659129241249171.

Lee, Barrett A., and Gregory Sharp. 2017. "Ethnoracial Diversity Across the Rural-Urban Continuum." *The ANNALS of the American Academy of Political and Social Science* 672

(1): 26–45. https://doi.org/10.1177/0002716217708560.

McMahon, Nicole, Anthony Sayers, and Christopher Alcantara. 2023. "Political Donations and the Gender Gap During COVID-19." *Party Politics* 29 (1): 176–84. https://doi.org/10.1177/13540688211047768.

McMahon, Tamsin, and Julia Wolfe. 2015. "Follow the Money: Mapping Canada's Richest Regions for Political Donations." In *The Globe and Mail*. https://www.theglobeandmail.com/news/politics/follow-the-money-mapping-canadas-richest-regions-for-political-donations/article26822168/.

Nathan, Charles, Arvind Krishnamurthy, Curtis Bram, and Jason Douglas Todd. 2025. "The Donor Went Down to Georgia: Out-of-District Donations and Rivalrous Representation." *Political Behavior* 47 (1): 41–60. https://doi.org/10.1007/s11109-024-09940-y.

Peoples, Clayton D., and Michael Gortari. 2008. "The Impact of Campaign Contributions on Policymaking in The U.S. And Canada: Theoretical and Public Policy Implications." In *Politics and Public Policy*, edited by Harland Prechel, vol. 17. Emerald Group Publishing Limited. https://doi.org/10.1016/S0895-9935(08)17003-6.

Pruysers, Scott. 2024. "Sour Grapes? Party Donors and Canadian Leadership Primaries." *Government and Opposition* 59 (2): 566–81. https://doi.org/10.1017/gov.2023.15.

Quillian, Lincoln. 2012. "Segregation and Poverty Concentration: The Role of Three Segregations." *American Sociological Review* 77 (3): 354–79. https://doi.org/10.1177/0003122412447793.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.

Sievert, Joel, and Stephanie Mathiasen. 2023. "Out-of-State Donors and Nationalized Politics in U.S. Senate Elections." *The Forum* 21 (2): 309–28. https://doi.org/10.1515/for-2023-2018.

Simas, Alexandre B., Wagner Barreto-Souza, and Andréa V. Rocha. 2010. "Improved Estimators for a General Class of Beta Regression Models." *Computational Statistics & Amp; Data Analysis* 54 (2): 348–66. https://doi.org/10.1016/j.csda.2009.08.017.

Statistics Canada. 2021. *Dissemination Area (DA)*. Dictionary, Census of Population, 2021. https://www12.statcan.gc.ca/census-recensement/2021/ref/dict/az/definition-eng.cfm?ID=geo021.

Statistics Canada. 2022a. *Boundary Files, 2021 Census*. Catalogue no. 92-160-X. Government of Canada. https://www12.statcan.gc.ca/census-recensement/2021/geo/sip-pis/boundary-limites/index2021-eng.cfm?year=21.

Statistics Canada. 2022b. *Census Profile, 2021 Census of Population – Download Files*. https://www12.statcan.gc.ca/census-recensement/2021/dp-pd/prof/details/download-telecharger.cfm?Lang=E.

Statistics Canada. 2025. *Postal Code Conversion File, June 2024, Census of Canada 2021*. Version V2. Borealis. https://doi.org/10.5683/SP3/IT8RET.

Tolley, Erin, Randy Besco, and Semra Sevi. 2022. "Who Controls the Purse Strings? A Longitudinal Study of Gender and Donations in Canadian Politics." *Politics & Gender* 18 (1): 244–72. https://doi.org/10.1017/S1743923X20000276.

Treasury Board of Canada Secretariat. 2026. *Open Government*. https://open.canada.ca/en.

Walks, R. Alan, and Larry S. Bourne. 2006. "Ghettos in Canada's Cities? Racial Segregation, Ethnic Enclaves and Poverty Concentration in Canadian Urban Areas." *Canadian Geographies / Géographies Canadiennes* 50 (3): 273–97. https://doi.org/10.1111/j.1541-0064.2006.00142.x.

Young, Lisa, and Harold J. Jansen, eds. 2011. *Money, Politics, and Democracy: Canada's Party Finance Reforms*. University of British Columbia Press. https://doi.org/10.59962/9780774818933.